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Two-stage approach for earthquake detection using multiple clustering-based classification

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SUMMARY

Deep learning (DL) approach has gained attention for earthquake (EQ) detection. To alleviate the problem of training data shortage, transfer learning (TL) provides a useful framework to adapt pre-trained models, typically through tuning of model parameters. Nonetheless, the current practice still requires considerable data, which hinders its application where only a small number of data is available. Instead of TL, we propose a novel two-stage of model correction as a solution to this important and ubiquitous problem in EQ detection. In the proposed approach, a pre-trained DL model is directly applied to waveform data in the target domain (first stage), and the cases that are classified as an earthquake signal (i.e., positive cases) are further classified as positives and negatives using a non-DL classification method (second stage). Our classification method for the second stage is based on multiple clustering, which characterizes local waveform patterns in terms of amplitude scale in specific time segments that are inferred in a data-driven manner. This characterization captures complex high-dimensional waveform patterns in a low-dimensional space, which leads to the effective classification of true and false positives. Furthermore, the proposed method is useful when only true positive waveforms are labeled

(PU classification). Both synthetic and real data analysis clearly demonstrated effectiveness of unsupervised waveform characterization of the proposed method.

Key words: Computational Seismology; Earthquake detection; Model correction; Deep learning.

1 INTRODUCTION

Recently, deep learning (DL) has gained considerable attention for earthquake (EQ) detection (Ross et al. 2018; Zhu & Beroza 2019; Mousavi et al. 2020; Katoh et al. 2025; Kaneko et al. 2023). Owing to the powerful representation capability of neural networks, a DL model is remarkably flexible for fitting to seismic waveforms (without confusion, seismic wave denotes an earthquake signal, hereafter). Constructing a DL model requires a large waveform dataset for high-performed seismic detection (Münchmeyer et al. 2022). However, large datasets may not be readily available or even exist for the target domain in question. To cope with the shortage of training data, transfer learning (TL) provides a useful framework to transfer knowledge from a data-abundant domain to a data-shortage domain (Yang et al. 2020; Zhuang et al. 2020; Weiss et al. 2016). Among the several TL approaches (Quinonero-Candela et al. 2008; Sugiyama et al. 2008; Dai et al. 2007; Pan et al. 2008), for earthquake detection, the model-based TL approach is widely used (Kim et al. 2023; Huynh et al. 2023; Vinard et al. 2022; Lapins et al. 2021; Lara et al. 2021; Chai et al. 2020), which aims to transfer knowledge of the classification model (Hinton et al. 2015; Frome et al. 2013). Typically, model-based TL entails re-training a pre-trained DL model using a relatively small dataset; this approach reduces the parameter search space by 'freezing' (i.e., fixing) parameters of some layers of the pre-trained model and/or initializing the model parameters with values from the pre-trained model (Goodfellow et al. 2016; Mesnil et al. 2012). In the context of EQ detection, the dataset size required for effectively optimizing a large number of model parameters remains large, which hinders the use of DL and TL for EQ detection when only small (dozens) seismic datasets are available, for instance, in frontier regions and sparsely instrumented regions and post-disaster response. Another direction of coping with labeled data shortage is self-supervised learning, which obtains representation of anchor instance (Jaiswal et al. 2020; Dosovitskiy et al. 2014). However, it is not straightforward to appropriately design neural architectures for encoding features and choose data augmentation method, which largely affect classification task (Jaiswal et al. 2020). Further, other state-of-the-art methods are based on foundation models (Si et al. 2024; Liu et al. 2024). A foundation model is a type of artificial intelligence (AI) model that is trained on a massive and diverse dataset. This extensive pre-training allows the model to learn a broad range of general patterns, structures, and relationships within the data, making it highly versatile. To apply a foundation model to earthquake detection, a number of labeled waveforms are still used, for examples, more than 1000 for fine-tuning (Liu et al. 2024).

The objective of the present study is to address the problem of data shortage for performing TL in the EQ detection model. We propose a novel model correction approach that is applicable when the number of available waveform data is much smaller than that is required by the aforementioned state-of-art methods. The key idea of our method is that instead of re-training a pre-trained model, we consider a model correction method that can be effectively constructed using a limited number of waveform data. We overcome a difficulty of classifying waveforms, which depends on dynamically changing amplitude patterns. The proposed method consists of two stages (Fig. 1). In the first stage, a pre-trained model is applied to waveform instances to classify them as seismic (i.e., positive cases) or non-seismic (i.e., negative cases). In the second stage, the positive cases are re-classified as positives and negatives using a non-DL method. This study contributes by proposing an effective classification model for the second stage. Specifically, we capture local waveform patterns as dynamically changing state of waveform, which is characterized by seismic wave amplitudes in several segments of datapoints. To perform such a local characterization of waveform, we build a multiple clustering model for waveform training, which simultaneously divides datapoints into several time segments and cluster waveform instances for each time segment by applying the method in Tokuda et al. (2017). Thus, the dynamically changing amplitude of a waveform instance is well characterized by its multiple cluster memberships in those time segments. Based on the similarity of multiple cluster memberships to training data, a new waveform instance is classified as positive or negative. Importantly, this second-stage method is readily applicable to only positive and unlabeled (PU) data as well, by restricting the similarity only to those labeled training data. Further, one may use positive waveform in a non-target domain, when labeled positive waveform is not available. In contrast with the proposed method, conventional supervised classification methods have difficulties of both capturing dynamically changing amplitude patterns of waveform and applying to PU data. In this manner, the proposed method provides a useful framework for EQ detection in cases of limited EQ waveform data.

2 METHOD

We propose a novel model correction method to effectively classify a waveform as earthquake or noise. Here, for simplicity, we focus on seismic P-wave, but it is applicable for S-wave as well.

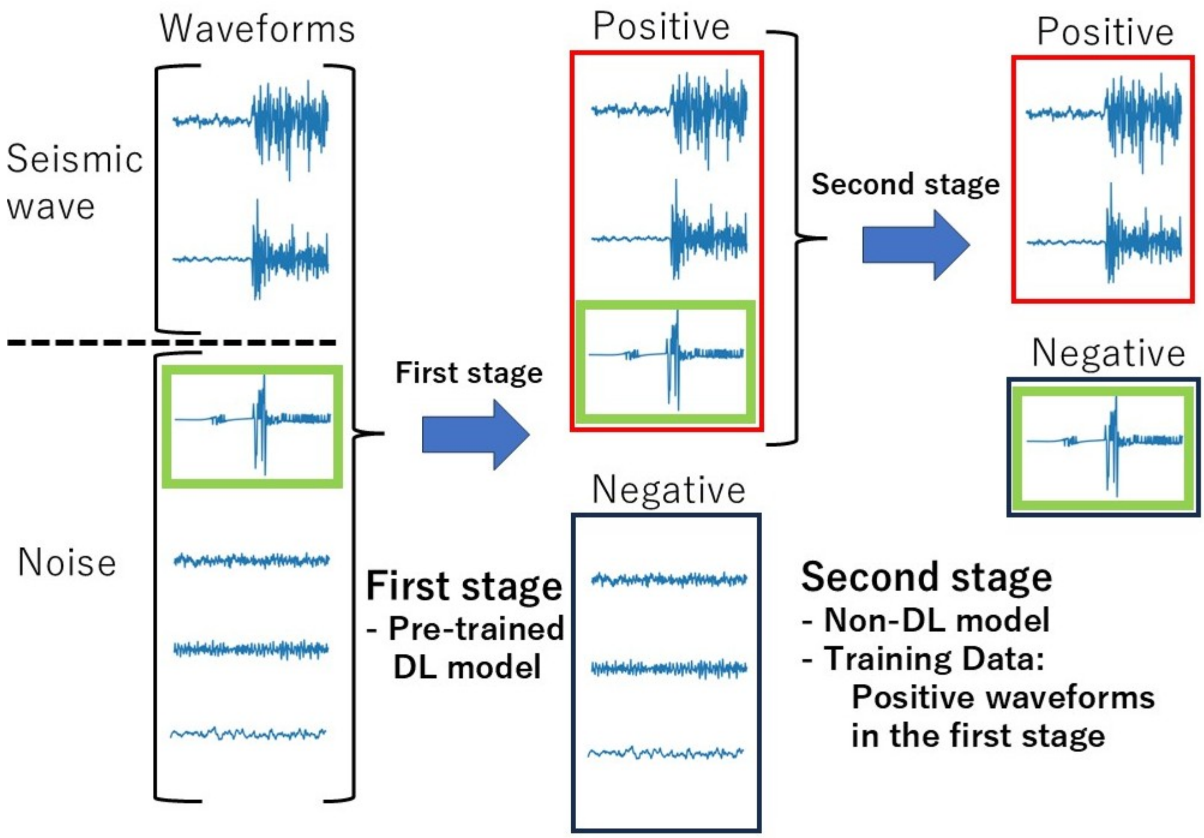


Figure 1. Illustration of the proposed two-stage method. In the first stage, waveforms are classified into 'positive' (earthquake) and 'negative' (noise) using a pre-trained deep learning (DL) model. In the second stage, those waveforms that have been classified as positive in the first stage are re-classified as positives and negatives. For waveform classification in the second stage, we use a non-DL method based on multiple clustering (Section 2.3), which does not require a large number of instances for model training. We build the second-stage classification model using the positively classified waveforms from the first stage and their label information on either positive (i.e., true positive in the first stage) and negative (i.e., false positive in the first stage). In this illustration, the noise waveform enclosed in the green rectangle was misclassified as positive in the first stage, but correctly classified as negative in the second stage. These waveforms were extracted from STEAD data (see Section 3.1).

2.1 Two-stage approach

The proposed method consists of two stages of classification, illustrated in Fig. 1. The first stage uses a pre-trained DL model trained using waveform data in the source domain (the source data domain may or may not be the same as the target domain). In the second stage, a non-DL model is applied to reclassify the positively classified waveforms from the first stage. The first-stage pre-trained model may be readily available, and using a small number of waveforms of the target domain, the second-stage classifier improves upon the classification accuracy of the first-stage classifier. We apply the second stage classifier only to the positively classified cases from the first stage. Correcting the false negatives

Algorithm 1 Multiple Clustering-Based Score (MCS)

Input: Training data $\mathbf{X}$; sample size of training data N; number of true positive (TP) n_t; number of false positive (FP) n_f; test instance $\mathbf{x}_{new}$

Train multiple clustering model:

  Function ϕ in Eq.(3), number of segments $V \leftarrow$ Multiple-clustering $(\mathbf{X})$

Characterization of training instances: $\boldsymbol{c}_i \leftarrow \phi(\boldsymbol{x}_i)$ for $i = 1, \dots, N$

Characterization of test instance: $\boldsymbol{c}_{new} \leftarrow \phi(\boldsymbol{x}_{new})$

Evaluation of MCS:

  MCS $\leftarrow \frac{1}{V \times n_t} \sum_{i \in TP} \sum_{v=1}^{V} \mathbb{I}(c_{new,v} = c_{i,v}) - \frac{1}{V \times n_f} \sum_{i \in FP} \sum_{v=1}^{V} \mathbb{I}(c_{new,v} = c_{i,v})$

Output: MCS

among the negatively classified cases from the first stage would be challenging, because of the large sample space of noise waveforms. Furthermore, false negatives implicitly affect the performance of the second stage, as they are excluded from the second-stage modeling. However, for simplicity, we do not evaluate such an impact of false negatives on the second-stage.

2.2 Pre-trained model for the first stage

In the first stage, we apply a pre-trained DL model to waveforms in the target domain. Here, we consider the recently developed 'GL model' in Tokuda & Nagao (2023). The GL model is an extension of generalized phase detection (GPD) model in Ross et al. (2018), which consists of three sub-models, capturing both global and local representations of waveforms. The reason for choosing the GL model is as follows. Well-performing deep learning models for earthquake detection include GPD, PhaseNet, and EQTransformer (Münchmeyer et al. 2022). GPD independently detects P- and S-phases by shifting 4-s time windows in 0.01-s increments, whereas PhaseNet and EQTransformer use longer time windows, aiming to simultaneously capture both phases within a single window. Consequently, in PhaseNet and EQTransformer, P- and S-phase detections are intertwined, implying that the detection of the P-phase may be influenced by that of the S-phase. To avoid such entanglement, the GPD-type model is preferable in our framework that focuses on a particular phase (P-phase in the present paper). However, it is well known that GPD produces a considerable number of false positives (Zhao et al. 2022). In contrast, the GL model suppresses false positives (Tokuda & Nagao 2023), making it well suited to the framework of our proposed approach.

For the GL model, each sub-model has the same architecture of the neural network as GPD: four convolutional and two fully connected networks. The final layer yields probabilities of P-wave, S-wave, and noise. For the GL model, those probabilities yielded by the three sub-models are combined as a product (i.e., point-wise multiplication). The model input is 4-s waveform data with three components

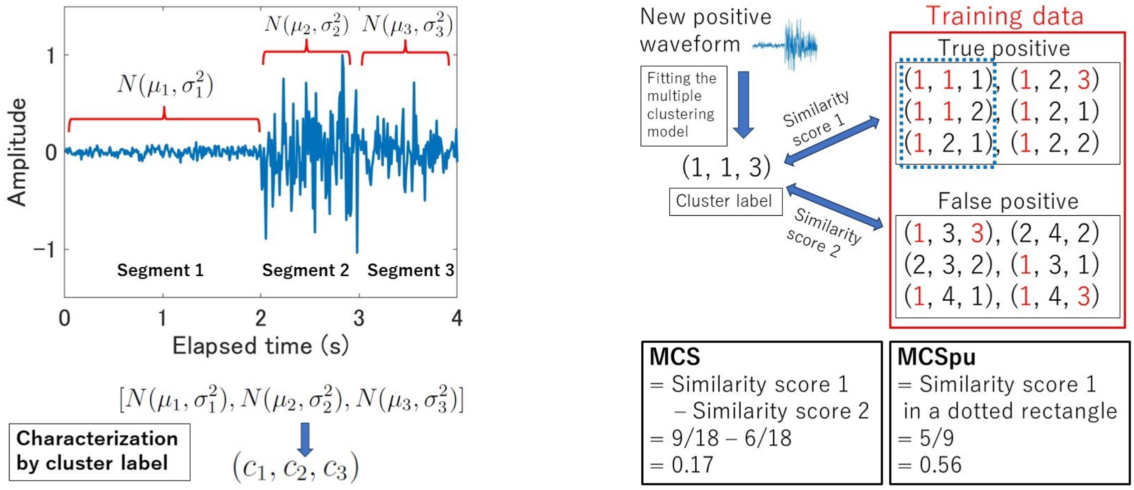
Algorithm 2 Multiple Clustering-Based Score for PU learning (MCSpu)

1: **Input:** Training data $\mathbf{X}$; sample size of training data N; number of labeled positive instances (LP) n_l; test instance $\mathbf{x}_{new}$
2: **Train multiple clustering model:**
3: Function ϕ in Eq.(3), number of segments $V \leftarrow$ Multiple-clustering ($\mathbf{X}$)
4: **Characterization of training instances:** $\boldsymbol{c}_i \leftarrow \phi(\boldsymbol{x}_i)$ for $i = 1, \ldots, N$
5: **Characterization of test instance:** $\boldsymbol{c}_{new} \leftarrow \phi(\boldsymbol{x}_{new})$
6: **Evaluation of MCSpu:**
7: MCSpu $\leftarrow \frac{1}{V \times n_l} \sum_{i \in LP} \sum_{v=1}^{V} \mathbb{I}(c_{new,v} = c_{i,v})$
8: **Output:** MCSpu

sampled at a frequency of 100 Hz. For continuous waveform data, a 4-s time window is shifted, e.g., by 0.01 s, seismic detection being evaluated per shift. Therefore, it does not pose a problem for shift invariance in determining the onset time. Further, in the simulation study by Tokuda & Nagao (2023), the mean absolute error of P- and S-picks are about 0.2 s. Hence, without possible negative influence, one may increase the window shift (for instance 0.05 s) for computational efficiency. We used the GL model that was trained using Southern California Seismic Network data (SCSN) (Tokuda & Nagao 2023). Importantly, the onset time of seismic phase in the training data is aligned at the center of the 4-s time window (i.e., at the time point of 2 s). Hence, the GL model gives high probability for a new seismic waveform when the onset of seismic phase coincides with the center of the 4-s time window. The alignment on the onset time allows for meaningful segmentation of datapoints over different waveforms in the second stage. Therefore, the performance of the proposed method could potentially deteriorate if there is no clear phase alignment of the onset time, for example, in the case of an emergent P-phase. However, for simplicity, we do not consider such nuisance cases in the present study.

2.3 Proposed method for the second stage

In the second stage, we correct the classification results of the first stage, focusing on positive cases. The overall scheme of the second stage is illustrated in Fig. 2. We built a classification model to discriminate between true and false positives, using waveforms in the target domain. For classification modeling, we hypothesize that the waveform differences between true and false positives are mainly due to amplitude patterns in specific segments of datapoints (e.g., Fig. 3). Based on this hypothesis, we captured waveform patterns by characterizing wave amplitudes in several segments of datapoints. For instance, in Fig. 2a, suppose that the waveform in the panel is well fitted by three Gaussian distributions, reflecting the underlying homogeneous distributions of amplitudes in each segment (Segments 1–3). We characterize the waveform in each segment by the fitted Gaussian distributions,



(a) Waveform characterization (b) Classification score

Figure 2. Illustration of the second-stage method. (a) Waveform characterization in terms of amplitude scale by cluster labels. (b) Example of score metric MCS in Eq.(4) and MCSpu in Eq.(5). First, a new waveform is characterized by the cluster label yielded by the multiple clustering model. Second, similarity scores 1 and 2 are evaluated as the mean agreement of cluster labels to true and false positives of the training data, respectively (cluster labels highlighted in red show agreement to the cluster label of the new instance). Lastly, MCS is evaluated by the difference between similarity scores 1 and 2. If we assume a PU situation in which labeled data are limited to true positives in the dotted rectangle, MCSpu is evaluated using similarity score 1 for those labeled data.

denoted as $N(\mu_1, \sigma_1^2), N(\mu_2, \sigma_2^2)$ and $N(\mu_3, \sigma_3^2)$, respectively. Here, $N(\mu, \sigma^2)$ denotes a Gaussian distribution with mean μ and variances σ^2. More simply, we can characterize the waveform using the corresponding cluster label (c_1, c_2, c_3) based on Gaussian mixture model clustering. Based on this characterization of waveform patterns, we built a classification model to discriminate between true and false positives (Fig. 2b). To this end, we propose the following multiple clustering-based method. The multiple clustering method infers the optimal segments and identifies overall fitted Gaussian distributions for all waveforms in data, which is equivalent to clustering waveforms with Gaussian mixture models for each segment (fitting a univariate Gaussian distribution to multivariate features). In sum, first, we characterize waveforms using multiple clustering, which captures waveform patterns. Second, based on the waveform characterization, we build a classifier between true and false positives. Unlike clustering methods based neural networks (Seydoux et al. 2020; Mousavi et al. 2019b), the multiple clustering method does not require a large number of instances for model building. Details of the second-stage method are provided in the following subsections.

2.3.1 Waveform characterization by multiple clustering

For waveform characterization, we consider a function $\phi(\cdot)$, which maps d-dimensional waveform (d is the number of datapoints) into a V-dimensional integer vector ($V \ll d$), $\phi : \mathbb{R}^d \to \mathbb{Z}^V$, where $\mathbb{R}$ is the set of real numbers; $\mathbb{Z}$ is the set of positive integers; and d and V are positive integers. Our motivation of introducing the function ϕ is to capture a waveform pattern as dynamically changing state of waveform in a low dimensional space, which provides a useful feature vector for discrimination between true and false positives.

Specifically, we define the function ϕ based on a multiple clustering method (Tokuda et al. 2017, 2018), which characterizes a waveform in terms of amplitude scale in a segment-wise manner (for a general account of multiple clustering, refer to Chapter 25 in Murphy (2012) and Yu et al. (2024)). In the context of EQ detection, the fundamental idea of using multiple clustering is that datapoints with similar amplitudes of waveform may be grouped into a single entity (segment) and fitted using a univariate Gaussian distribution (Fig. 2a), which introduces segment-wise cluster structure of waveform instances (i.e., modeling Gaussian mixtures in each segment). That is, for the $n \times d$ waveform data matrix $\boldsymbol{X} = \{\boldsymbol{x}_i\}$ ($\boldsymbol{x}_i$ denotes waveform instance i ($1 \leq i \leq n$)), we consider the following log-likelihood (the base is Napier constant):

$$\log P(\boldsymbol{X}|\boldsymbol{Y}, \boldsymbol{Z}, \boldsymbol{\Theta}) = \sum_{i=1}^{n}\sum_{j=1}^{d}\sum_{v=1}^{V}\sum_{k=1}^{K_v} \mathbb{I}(Y_{j,v} = 1)\mathbb{I}(Z_{i,v,k} = 1) \log N(x_{i,j}|\mu_{v,k}, \sigma_{v,k}^2), \tag{1}$$

where K_v is the number of instance clusters for segment v; $Y_{j,v}$ is an indicator (0 or 1) of whether datapoint j belongs to segment v with $\Sigma_v Y_{j,v} = 1$; $Z_{i,v,k}$ is an indicator of whether instance i belongs to cluster k in segment v with $\Sigma_k Z_{i,v,k} = 1$; $\mu_{v,k}$ and $\sigma_{v,k}$ are the mean and standard deviation for cluster k in segment v, respectively; $\boldsymbol{\Theta}$ is a collection of all relevant parameters; $N(\cdot|\mu, \sigma^2)$ is a Gaussian distribution with mean μ and standard deviation σ; and $\mathbb{I}(\cdot)$ is an indicator function ($\mathbb{I}(a) = 1$ if the statement a is true; 0 otherwise). Both the segment and cluster labels are denoted as integers. A more general form is to model cluster structure of datapoints within a segment as well (i.e., co-clustering of instances and datapoints), which is formulated as

$$\log P(\boldsymbol{X}|\boldsymbol{Y}, \boldsymbol{Z}, \boldsymbol{\Theta}) = \sum_{i=1}^{n}\sum_{j=1}^{d}\sum_{v=1}^{V}\sum_{k=1}^{K_v}\sum_{g=1}^{G_v} \mathbb{I}(Y_{j,v,g} = 1)\mathbb{I}(Z_{i,v,k} = 1) \log N(x_{i,j}|\mu_{v,k,g}, \sigma_{v,k,g}^2), \tag{2}$$

where G_v is the number of datatpoint clusters for segment v; $Y_{j,v,g}$ is an indicator of whether datapoint j belongs to cluster g within segment v with $\Sigma_{v,g} Y_{j,v,g} = 1$. $\mu_{v,k,g}$ and $\sigma_{v,k,g}$ are respectively the mean and standard deviation for instance cluster k and datapoint cluster g in segment v. The indicator tensors $\boldsymbol{Y}$ and $\boldsymbol{Z}$ are randomly initialized and inferred in the Bayesian framework (for details, refer to Tokuda et al. (2017)). The standard deviation (s) of Gaussian distribution corresponds to the average amplitude (s) of waveform in a segment, which locally characterizes the waveform pattern. This leads to characterization of a waveform in a particular segment by its cluster membership. The multiple clustering algorithm (Tokuda et al. 2017) reveals such a data structure by simultaneously identifying optimal segmentation and segment-wise co-clustering of waveforms and datapoints, which yields a multiple clustering model. Applying the multiple clustering model to waveform data, we define the function $\phi(\cdot)$ for a waveform $\boldsymbol{x} \in \mathbb{R}^d$:

$$\phi(\boldsymbol{x}) = (c_1, \dots, c_V), \tag{3}$$

where $c_v \in \mathbb{Z}$ $(1 \leq v \leq V)$ denotes a cluster membership of the waveform $\boldsymbol{x}$ for segment v by the multiple clustering model. The number of segments V and clusters of instances and datapoints in each segment are automatically inferred in a data-driven manner using Dirichlet process (Gelman et al. 2013). That is, one does not need specify those numbers in advance (we set the hyperparameter of Dirichlet process, i.e., concentration parameter α, to one). For a new waveform instance, we evaluate its cluster memberships by fitting the multiple clustering model obtained from the training data. For computational efficiency, we set the maximum number of V to five in the present paper. The computational complexity of the algorithm is $O(nd)$ where n and d are sample size and datapoints, respectively (Tokuda et al. 2017).

To characterize waveforms, one may wonder to use other unsupervised methods such as principal component analysis (PCA), uniform manifold approximation and projection (UMAP) and convolutional autoencoders. These methods map high-dimensional waveforms into a low-dimensional space, which is useful for feature extraction. However, the relevance of these extracted features to our classification task is not guaranteed. In contrast, our multiple clustering method is explicitly based on the assumption that earthquake and noise exhibit different amplitude patterns, which motivated our choice of this method.

2.3.2 Classifier

We build a classification model based on the waveform characterization in Eq.(3). Let us assume n_t true and n_f false positive cases in the training data. We denote the waveform characterization of a new waveform $\boldsymbol{x}_{new}$ as $(c_{new,1}, \dots, c_{new,V})$. For this new instance, we quantify the degree of similarity to true and false positive cases in the training data. To this end, we define ‘multiple clustering-based score’ (MCS) based on the mean agreement of cluster memberships, as follows:

$$\mathrm{MCS}(\boldsymbol{x}_{new}) = \frac{1}{V \times n_t} \sum_{i \in TP} \sum_{v=1}^{V} \mathbb{I}(c_{new,v} = c_{i,v}) - \frac{1}{V \times n_f} \sum_{i \in FP} \sum_{v=1}^{V} \mathbb{I}(c_{new,v} = c_{i,v}), \tag{4}$$

where TP and FP denote the set of waveforms of true and false positive cases in the training data, respectively. The first term on the right-hand side of Eq.(4) evaluates the mean agreement of cluster memberships to true positive cases over segments, whereas the second term evaluates the mean agreement to false positive cases (refer to Fig. 2b). Since MCS is defined as the difference between these two terms, the waveform is more likely to be a true positive if MCS is larger than zero, and a false positive otherwise. In practice, the MCS threshold may be given by the practitioner depending on practical purposes.

We extend the definition of MCS to label and unlabeled data. As a natural extension, we simply restrict sets TP and FP to those labeled instances. Note that the multiple clustering model does not require labeled data, hence, the cluster memberships of the training data are not affected by the label status on instances. In particular, we consider PU classification in which only some positive instances are labeled (Elkan & Noto 2008; Bekker & Davis 2020). Denoting the set of labeled positive instances as LP, for PU classification, we define 'multiple clustering-based score for PU learning' (MCSpu) as follows:

$$\mathrm{MCSpu}(\boldsymbol{x}_{new}) = \frac{1}{V \times n_l} \sum_{i \in LP} \sum_{v=1}^{V} \mathbb{I}(c_{new,v} = c_{i,v}), \tag{5}$$

where n_l is the sample size of LP. Unlike MCS, MCSpu has no clear threshold to determine true and false positives. Hence, the threshold is provided by the practitioner. The pseudocodes for MCS and MCSpu are provided in Algorithms 1 and 2, respectively.

2.4 Experiment of the proposed second stage method

We conducted an experiment to evaluate the performance of the proposed second stage using synthetic data. We generated surrogate datasets for true and false positives, considering the following two-way experimental design. One factor is the cluster structure of false positives (one modal/two modals) in which we evaluate performance for single/multiple patterns of false positives. The second factor is the temporal pattern of amplitudes (discrete change/continuous change) in which we evaluate performance when amplitude patterns follow/do not follow the assumption of our model (i.e., the underlying amplitudes are not constant in a segment). For simplicity, we assumed linear decay of amplitude for both the true and false positive waveforms. In addition, we performed a supplementary analysis, assuming exponential decay of the true positive waveforms (Aki & Chouet 1975). For these two factors, we generated synthetic datasets. We applied the second-stage multiple clustering-based classifier to the synthetic dataset and evaluated its performance to discriminate between true and false positives by computing the area under the curve (AUC) of the receive operation characteristic curve (Fawcett 2006). We compared the performance of the proposed method with those of the baseline classification methods. Experiments (including those in Section 3) were run on a computer with an Intel Xeon(R) Gold 6348 CPU running at 2.60 GHz, running AlmaLinux version 8.5 (Arctic Sphynx).

2.4.1 Data generation

We generated synthetic waveform (referred to as 'Toy data') by sampling from Gaussian distributions. Using Gaussian models may be gross simplification of mimicking real waveform, compared with the state-of-the-art method such as generative adversarial network model (Wang et al. 2021). Further, in a real situation, there is more complexity in waveforms, because it may include coda, scattering and unknown noise. However, the use of Gaussian-based toy data has an advantage of easily controlling feature parameters, which facilitates to evaluate performance. We assume that the waveform duration is 4 s with a sampling rate of 100 Hz and composed of three components (we do not discriminate between up-down, east-west, and south-north components), the amplitude of each datapoint being independently sampled from a Gaussian distribution $N(0, \sigma^2)$ (mean 0, standard deviation σ) in a segment-wise manner. As shown in Fig.2a, we assume three segments 1–3 that comprise datapoints of durations 0–2 s, 2–3 s, and 3–4 s, respectively, with $\boldsymbol{\sigma} = (\sigma_1, \sigma_2, \sigma_3)$ being the corresponding standard deviation of Gaussian distribution. For Toy data 1, we set $\boldsymbol{\sigma} = (0.1, 1, 0.5)$ for true positive waveform, and $\boldsymbol{\sigma} = (0.1, 1, 0.6)$ for false positive waveform. In this setting, the difference between true and false positive waveforms is only between 3 s and 4 s. Hence, by design, both true/false positive waveforms are very similar. For Toy data 2, true positive waveform was sampled in the same manner as in Toy data 1. However, for false positive waveform, two modalities were considered. One half of the instances were sampled as in Toy data 1, and the other half were sampled from $\boldsymbol{\sigma} = (0.1, 1, 0.3)$. Note that the amplitude of true positive waveform in segment 3 is between these two modalities of false positive waveform. The setting for Toy data 3 was similar to that of Toy data 1, but the amplitudes between 2 s and 4 s decreased linearly. That is, for true positive waveform, we set $\sigma = 1 - 0.3 \times (t - 2)$ at the time point t s ($2 \leq t \leq 4$), and $\sigma = 1.1 - 0.3 \times (t - 2)$ for false positive waveform. For Toy data 4, true positive waveform was sampled in the same manner as in Toy data 3. However, for false positive waveform, half of the instances were sampled as in Toy data 3, and the other half using $\sigma = 0.8 - 0.3 \times (t - 2)$. For training data, we randomly generated n (referred to as 'sample size per attribute', hereafter) instances for true and false positive cases, resulting in a dataset with sample size $2n$, where n was varied as 20, 50, 100, 200 and 500. For each training data, we generated test data with $n = 100$. We built a classification model using the training data, whereas the classification performance was evaluated using the test data. Toy data 1 and 2 coincide with the assumptions of the multiple clustering method in which the cluster structure is identified by fitting finite Gaussian mixture models. By contrast, Toy data 3 and 4 do not coincide with those assumptions, because the amplitudes between 2 s and 4 s changed continuously. As a supplementary setting, we considered exponential decay of amplitude between 2 s and 4 s (Toy data 5 and 6; for details, refer to the caption of Fig. S3) to model a more realistic true positive waveform (Aki & Chouet 1975). We vectorized the three-component

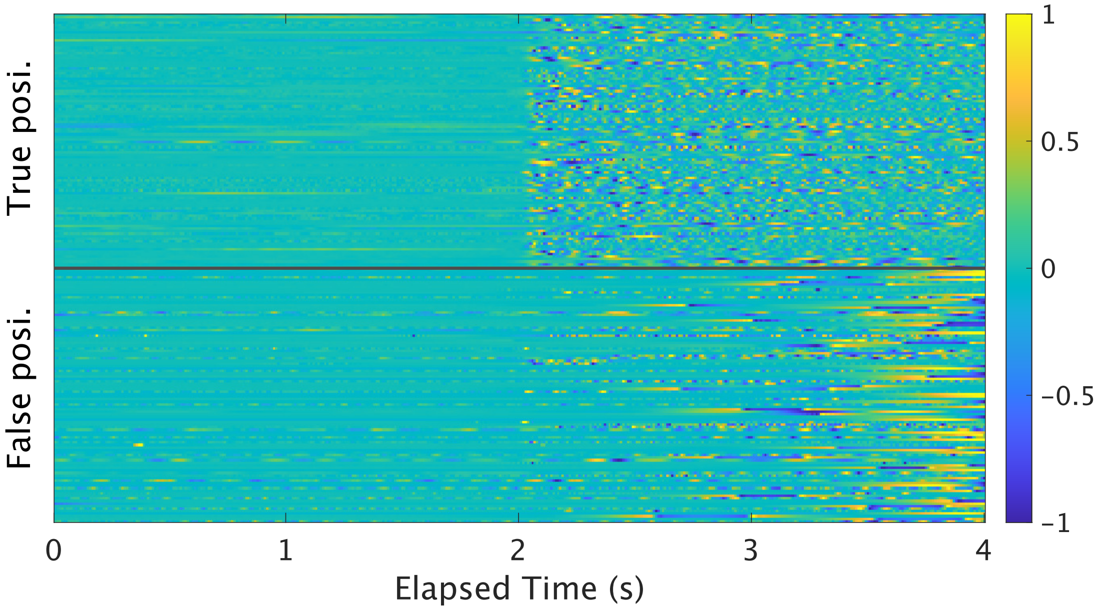
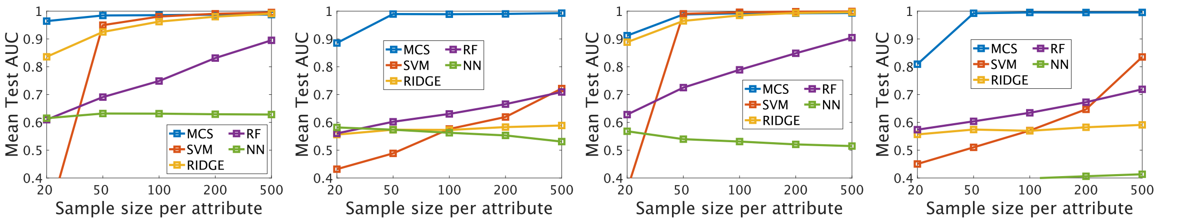


Figure 3. 4-s waveforms (up-down component) of STEAD data. 100 instances are displayed for true positive ('True posi.') and false positive ('False posi.') by the first stage method in Section 2.2. The color denotes amplitude.

waveform before inputting to the classification model. See waveform examples of these Toy data in Figs. S1 and S2.

2.4.2 Baseline classification methods

We compared the performance of the proposed method with other classification methods: support vector machine (SVM), ridge logistic regression (RIDGE), random forest (RF), and nearest neighbor (NN). These are widely-used linear (SVM, RIDGE) and nonlinear (RF, NN) classification methods. Since we evaluate the performance using AUC, we set that these classification methods yielded a continuous classification score. We used the Python implementations of these classifiers provided in the Python library 'scikit-learn'. The settings of these methods were as follows (for details about these methods, refer to Hastie et al. (2009)). For SVM, data were standardized (i.e., mean 0 and variance 1 for each datapoint). The radial basis function kernel was used in which the scale parameter γ was set as the reciprocal of the number of features with a probability output given by the Platt method (Platt et al. 1999). For RIDGE, data were standardized. The regularization parameter for L_2 norm was optimized among 0.001, 0.01, 0.1, 1, 10, 100 and 1000 using five-fold cross validation. For RF, the number of trees was set to 100. The Gini impurity was used to evaluate the loss function. The maximum number of depth was not set; hence, nodes were expanded until all leaves were pure. The classification score was evaluated as the mean of predicted class probabilities over 100 tress. For NN, to evaluate a new instance, five nearest instances in the training data were identified based on Euclidean distance. Subsequently, the classification score was evaluated as the proportion of the true positive cases over those five instances.



(a) Toy data 1 (b) Toy data 2 (c) Toy data 3 (d) Toy data 4

Figure 4. Classification performance of true and false positives for Toy data. (a) Toy data 1 (discrete, one modal). (b) Toy data 2 (discrete, two modals). (c) Toy data 3 (continuous, one modal). (d) Toy data 4 (continuous, two modals). MCS is the proposed method. For details of the results, refer to Tables S1–S4.

We took absolute values of the input data for these baseline methods (not for the proposed method), because our preliminary analysis suggested rather poor performance of the baseline methods otherwise (possibly due to the alternation of positive/negative amplitudes).

2.4.3 Results

We evaluated the mean AUC over 100 randomly generated samples for each sample size per attribute n of training data. The proposed method outperformed the baseline methods with the mean AUC being close to 1 for $n \geq 50$ (Fig. 4). This suggests the applicability of the proposed method when amplitude patterns do not follow the model assumption that waveform amplitude changes in a discrete manner (Fig. 4c, d). The baseline methods did not work well with AUC < 0.8 when false positives have two modalities (Fig. 4b, d). For the exponential decay waveform (Toy data 5 and 6), similar results were obtained (Fig. S3). Furthermore, unlike the baseline methods, the proposed method is not prone to overfitting when the training data size is small, with classification performance remaining almost identical between the training and test data (Figs. S4 and S5).

3 APPLICATION TO REAL DATA

We applied the proposed method to real waveform data. The main focus was on the performance of the second-stage method. First, we applied the proposed method to the Stanford Earthquake Dataset (STEAD), which is a global seismic dataset (Mousavi et al. 2019a) and often used as a benchmark for evaluating seismic detection methods (Münchmeyer et al. 2022; Mousavi et al. 2020). Second, we applied the proposed method to the transformed STEAD data in which we transformed the true positive waveform by reducing the discrepancy between true and false positives. Third, we applied the proposed method to the STEAD data assuming that only some positive labels were available in the context of PU classification. For all these datasets, we evaluated the performance in terms of AUC as described in Section 2.4. For the first stage, we used the GL model, described in Section 2.2. For these applications, the input instance was a vectorized 4-s waveform of three components with a sampling rate of 100 Hz, resulting in 1200 datapoints.

3.1 STEAD data

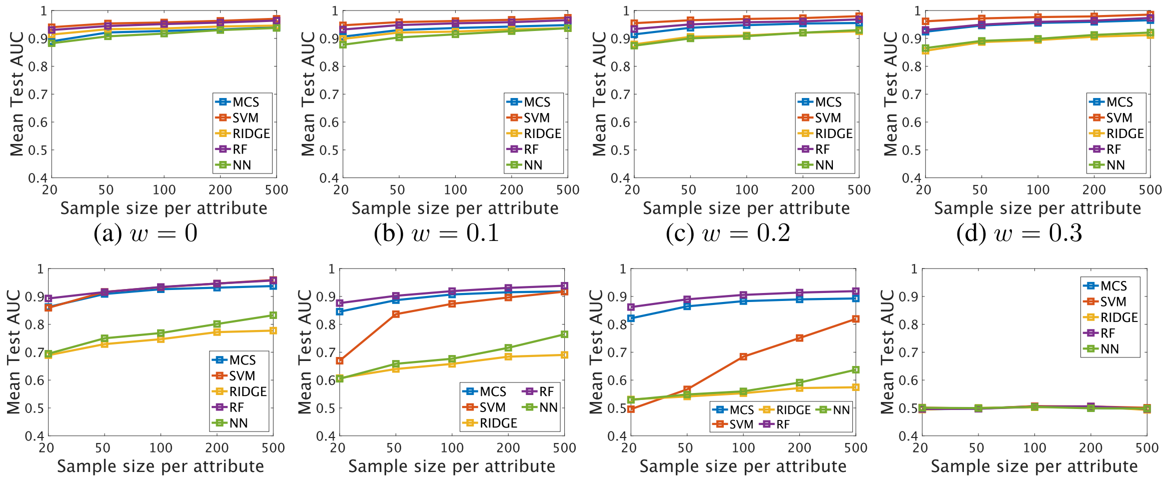
The original STEAD data consists of over 1 million instances of 60-s P-wave, S-wave, and noise waveforms. From this data, we extracted 4-s noise instances and applied the GL model with the probability threshold 0.5, which resulted in 13080 false positive waveforms for P-wave (i.e., noise instances, but erroneously identified as P-wave by the GL model). For the true positive waveforms, we extracted 4-s waveforms with the onset time of P-wave being centered. We applied the GL model to these instances and uniformly randomly selected 10000 waveforms that were detected as P-wave. We concatenated these true and false positive instances, which we refer to as 'STEAD data' hereafter. For training data, by varying the sample size per attribute n as 20, 50, 100, 200, and 500, we uniformly randomly selected instances for true and false positives, whereas for test data, the sample size was fixed to 100 for true/false positive. For each value of n, 100 pairs of training and test data were generated and the performance was evaluated using AUC. The results are presented in Fig. 5a. The mean AUC values for the proposed and baseline methods were high ($>$ 0.9) for all values of n. Although the differences in performance were small among these methods, SVM and RF performed slightly better. Overall, for the STEAD data, the true and false positive cases were well separated (Fig. 3), which facilitated accurate classification by all methods.

3.2 Transformed STEAD data

We evaluated the classification performance by manipulating the separability between true and false positives. To this end, we assumed that a true positive waveform may be superpositioned by a false positive waveform. To enforce this assumption, we took a weighted sum of the true and false positive waveforms, which can be seen as data augmentation (Zhu et al. 2020). Specifically, denoting true and false positive waveforms as $\boldsymbol{x}_{TP}$ and $\boldsymbol{x}_{FP}$, respectively, we transformed the true positive waveform as follows:

$$\boldsymbol{x}_{TP} \leftarrow w \times \boldsymbol{x}_{FP} + (1 - w) \times \boldsymbol{x}_{TP}, \tag{6}$$

where $\boldsymbol{x}_{FP}$ was uniformly randomly sampled from the STEAD data; w is the proportion of the false positive waveform that was overlaid on the true positive waveform. We varied w from 0 to 1 in increments of 0.1. For each value of w, we evaluated model performance as described in Section 3.1. The results are presented in Fig. 5. As w increased from 0 to 0.3, the performance of RIDGE and NN gradually deteriorated, whereas that of the proposed method, SVM, and RF remained the same. With



(a) $w = 0$ (b) $w = 0.1$ (c) $w = 0.2$ (d) $w = 0.3$

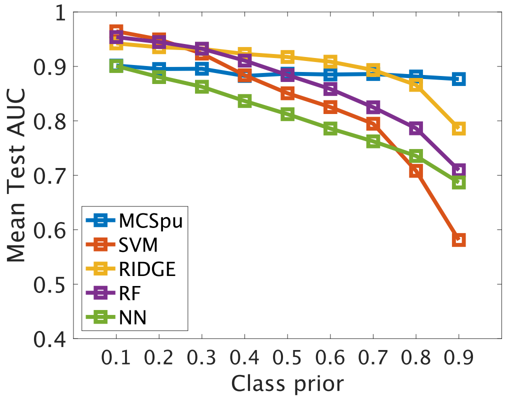
(e) $w = 0.7$ (f) $w = 0.8$ (g) $w = 0.9$ (h) $w = 1$

Figure 5. Classification performance of true and false positives for the transformed STEAD data. The weight w in Eq.(6) was varied from 0 to 1 in increments of 0.1 (for space constraints, panels with $w = 0.4, 0.5$, and 0.6 are not shown). MCS is the proposed method. The $w = 0$ case is equivalent to the setting of Section 3.1. For details of the results, refer to Tables S5–S12.

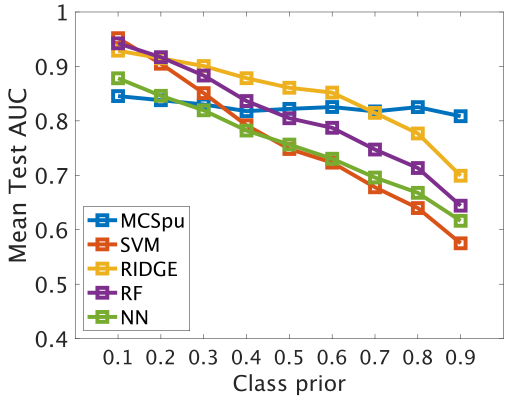
further increase in w from 0.7 to 0.9, the performance of RIDGE, NN, and SVM deteriorated, whereas the proposed method and RF performed very well with AUC > 0.8. As expected, all methods failed to discriminate between true and false positives at $w = 1$.

3.3 PU classification of STEAD data

We evaluated PU classification performance using STEAD data. Here, we assumed the *case-control scenario* in which the positive and unlabeled instances come from two independent datasets (Bekker & Davis 2020). In this experiment, we considered two cases for labeled instances: (1) STEAD data; (2) SCSN data. In case (2), non-target-domain instances are used, which would be an alternative scenario if target-domain instances were not readily available. For the sample size of the training data, we considered a similar setting as in Kwon et al. (2020), setting the labeled positive sample size and the unlabeled sample size to 100 and 400, respectively. We varied the class prior (i.e., the proportion of positive instances in unlabeled data) from 0.1 to 0.9. For the test data, we fixed the sample size to 100 for both true and false positives. For each value of class prior, we randomly generated 100 pairs of training and test data, which were evaluated using the mean AUC of test data. As regards classification algorithms, we built a classifier based on Eq.(5) for the proposed method, whereas for SVM, RIDGE, and RF (we did not change the setting of RF for imbalanced data), we used the biased learning approach (i.e., considering unlabeled instances as negative), which is a widely-used method in PU classification (Bekker & Davis 2020; Jain et al. 2017). In addition, we used another popular approach based on empirical risk minimization (ERM) (Jiang et al. 2023; Du Plessis et al. 2014). The ERM approach is



(a) STEAD labeled positive



(b) SCSN labeled positive

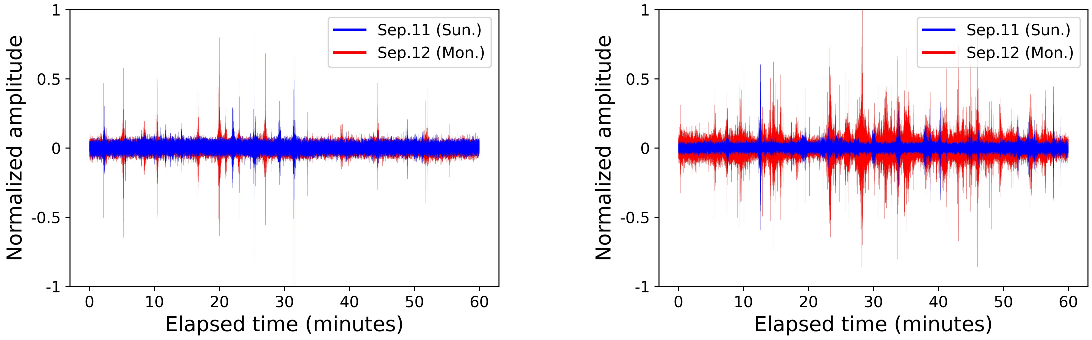
Figure 6. PU classification performance for STEAD data. (a) case (1), and (b) case (2). The proposed method was based on MCSpu given in Eq.(5) ('MCSpu'). SVM, RIDGE, and RF were based on the biased learning framework. For details of results, refer to Tables S13 and S14.

not applicable for RF, due to negative weighting. For NN, we simply considered those labeled instances for positive. The performance results are summarized in Fig. 6 and Fig. S6. In case (1), the proposed method (MCSpu) performed very well with AUC being about 0.9 over different class priors. The other methods performed very well for small class prior with AUC > 0.9, but the performance deteriorated as the class prior increased (e.g., AUC < 0.8 for class prior 0.9). Although a similar tendency was observed in case (2), the performance slightly worsened for all methods due to the domain difference between the training and test data.

3.4 MeSO-net data

Lastly, we applied the proposed method to continuous waveform data obtained from the Metropolitan Seismic Observation network (MeSO-net) in Japan (Aoi et al. 2021; NIED 2021). For the MeSO-net, 300 acceleration seismometers are installed in the highly populated Tokyo metropolitan areas with the interval of 2-5 km (installation depth 20 m). Because of the proximity to the urban area, the observed waveforms are vulnerable to city noise (Kawakita & Sakai 2009), which potentially causes false positive seismic detection. For this data, in particular, we focused on the period of Sep. 4-12, 2011, when a number of earthquakes were observed due to aftershocks of the 2011 Tohoku-Oki earthquake. The waveform data for 13 seismic stations in the Narita region during this period were intensively studied in Yano et al. (2021). In terms of type of seismometers and sampling rate, MeSO-net data (acceleration seismograph with sampling rate 200 Hz) is rather different from the SCSN data (velocity seismograph with sampling rate 100 Hz) that was used for training the GL model.

We constructed the training dataset using waveforms recorded during September 4–10, and the test dataset using waveforms from September 11–12. The application procedure of the proposed method was

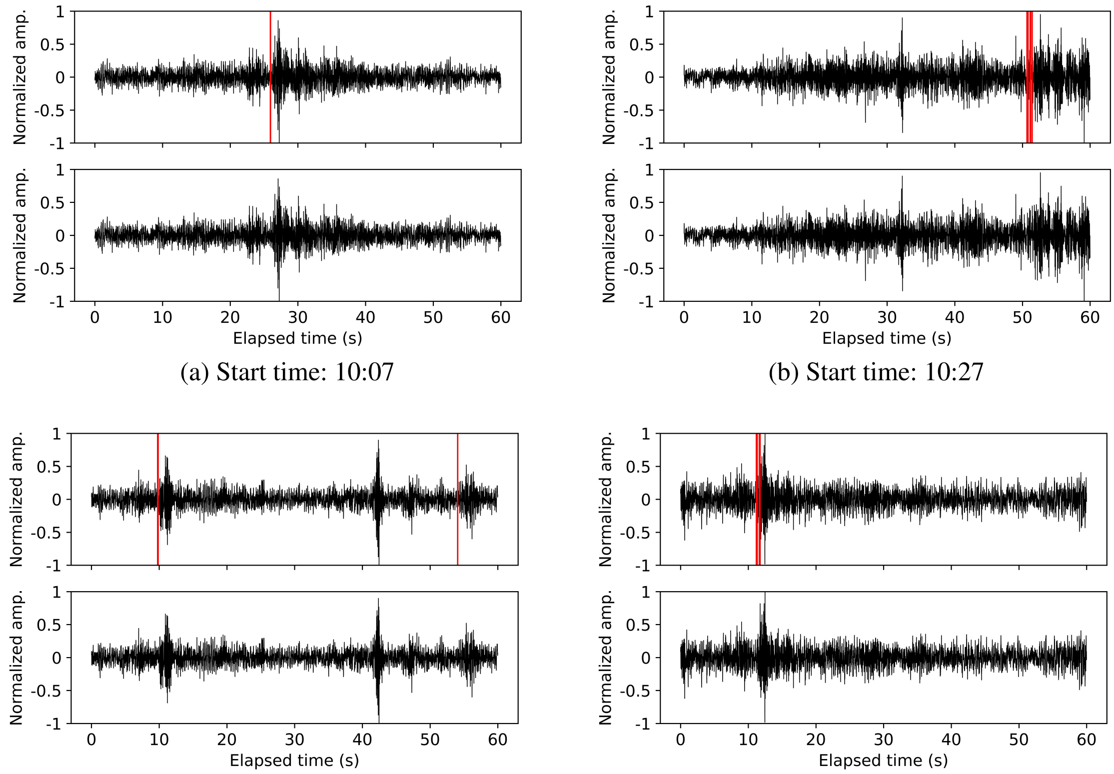


(a) Start time: 00:00 am (b) Start time: 10:00 am

Figure 7. One-hour waveform examples for Station MD1M (up-down component). Blue and red waveforms denote those on Sep.11 (Sunday) and Sep.12 (Monday), respectively. For panel (a), the time starts at 00:00 am, whereas for panel (b), it starts at 10:00 am. Both panels, amplitudes are normalized by subtracting the mean, followed by division by the maximum absolute amplitudes. Normalization is performed using waveforms on both Sep.11 and Sep.12 in each panel.

as follows. Focusing on station MD1M, we first downsampled the continuous waveform from 200 Hz to 100 Hz by decimation, and then applied the GL model to the decimated data using a 4-s time window with a 0.01-s time shift. Next, referring to the seismic catalog of MeSO-net, we excluded waveforms within five minutes before and after a cataloged event, and identified false-positive waveforms with a P-phase probability larger than 0.5. This yielded 24,732 false-positive waveforms (17,876 for training and 6,856 for test data), where multiple waveforms may correspond to a single event due to the small time shift. The main cause of these false positives was city noise, as suggested by Fig. 7, which shows large waveform amplitudes during working hours between 10:00 and 11:00. This, in turn, explains the difference in the number of false positives between 10:00 and 11:00 on Sunday (Sep. 11) and Monday (Sep. 12), with 171 and 1,465 cases, respectively. For true positive waveforms, we identified 114 (86 for training and 28 for test data) events with probability of P-phase > 0.5 for MD1M, whereas 166 events were false negative. Such a low detection rate was possibly due to the low signal-to-noise ratio (SNR) for seismic events (for SNR in MeSO-net data, please refer to Fig. S3 in Yano et al. (2021)).

We trained the second stage classification model, using randomly selected 80 true positive and 80 false positives from the training data. The value of AUC for the training data was 0.95. Next, we applied the model to the test data. Because of extremely imbalance of test data, we did not evaluate AUC. Rather, setting the MCS threshold to zero, we report on a confusion matrix in Table 1. Moreover, waveform examples of successful detection of false positives are displayed in Fig. 8, whereas those of failed cases are in Fig. 9.



(a) Start time: 10:07

(b) Start time: 10:27

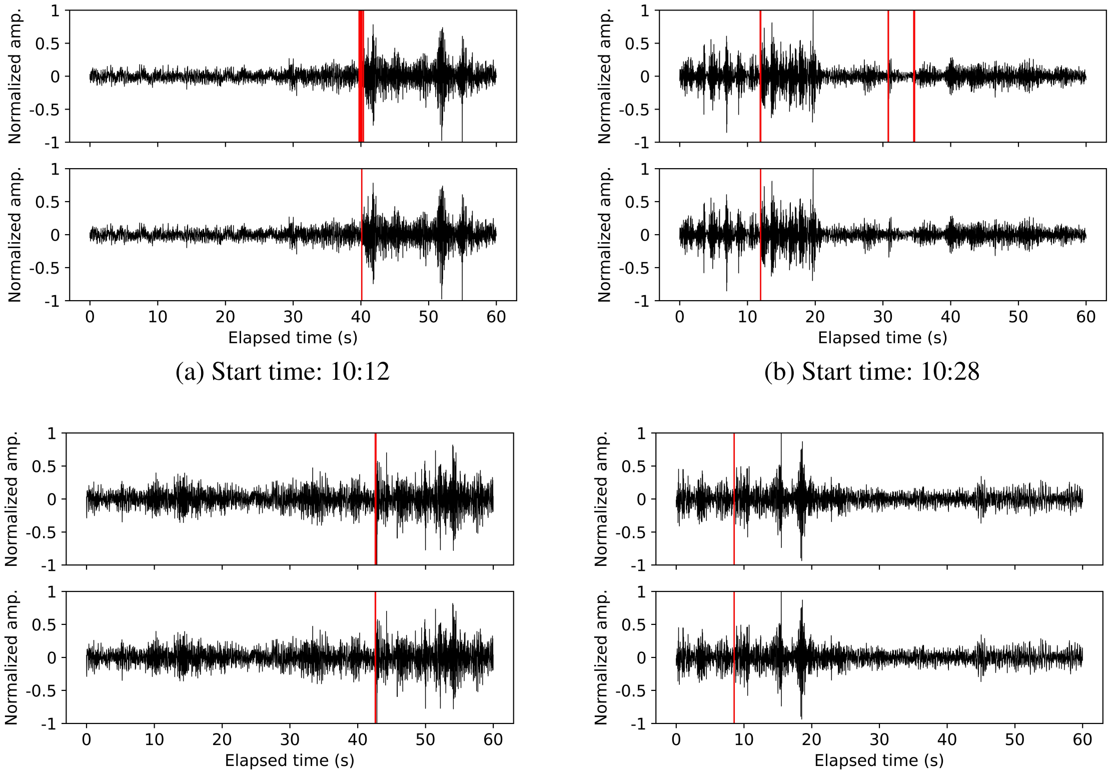
(c) Start time: 10:30

(d) Start time: 10:36

Figure 8. Examples of false positive waveforms of MeSO-net data (station MD1M) that were correctly identified by the proposed method in the second stage. The detection threshold for MCS was set to zero. All examples were obtained between 10:00 am and 11:00 am on 2011/09/13 with the start time being shown in the panel title. For the upper sub-panel of each example, the waveform of up-down component is shown where the red line denotes the P-phase picking (evaluated every 0.01 s) by the first stage method (i.e., GL model). For the lower sub-panel, the same waveform is shown, but the red lines were removed for those pickings that were evaluated as false positive by the proposed method.

Table 1. Results of performance by the proposed method for MeSO-net test data. The confusion matrix summarizes numbers of true and predicted labels. 'TP' denotes true positive whereas 'FP' denotes false positive.

		Predicted label		
		TP	FP	Total
True label	TP	24	4	28
	FP	584	6272	6856
	Total	608	6276	6884



(a) Start time: 10:12 (b) Start time: 10:28

(c) Start time: 10:31 (d) Start time: 10:41

Figure 9. Examples of false positive waveforms of MeSO-net data (station MD1M) that were not correctly identified by the proposed method in the second stage. The detection threshold for MCS was set to zero. All examples were obtained between 10:00 am and 11:00 am on 2011/09/13 with the start time being shown in the panel title. For the upper sub-panel of each example, the waveform of up-down component is shown where the red line denotes the P-phase picking by the first stage method (i.e., GL model). For the lower sub-panel, the same waveform is shown, but the red lines were removed for those pickings that were evaluated as false positive by the proposed method.

4 DISCUSSION

We developed a novel two-stage model correction approach for seismic detection. The distinctive feature of the proposed method is its applicability to waveform datasets with small sample sizes. The proposed second-stage method characterizes waveforms by amplitude scale based on multiple clustering, which has the potential to outperform conventional classification methods to discriminate between true and false positives. Moreover, the proposed method is not prone to overfitting, because it uses relatively small number of parameters by fitting a univariate Gaussian distribution to multiple features. Furthermore, the proposed method is applicable for PU situations. Unlike the benchmark methods, its performance was high irrespective of class prior, which has a practical advantage over the benchmark methods because class prior is not known in general. The main reason of insensitivity to class prior is that model-building of the proposed method is based on clustering without using label information, which is not influenced by the absence of negative labels. This suggests that the model does not suffer from spurious associations between the input data and the output resulting from the absence of negative labels in the PU setting. Moreover, unlike other supervised methods, the proposed approach enables estimation of test data classification performance from training data, owing to its non-overfitting nature.

The application of the proposed method to raw waveform streams from MeSO-net demonstrated its effectiveness in a real-data setting. Nonetheless, there were cases in which the method failed to correctly identify false positives (Fig. 9). These failed waveforms were most likely caused by city noise, although other waveforms generated by city noise were successfully identified (Fig. 8). This raises the question of whether unidentifiable false positives caused by city noise possess particular features, a question that is not straightforward to answer at this moment.

For the proposed method, threshold selection for MCS and MCSpu is important for tuning detection performance. For MCS, the threshold may be set to zero by default, or alternatively determined from the ROC curve of the training data (Fig. S7). Because the proposed method is less prone to overfitting, the ROC curve of the training data provides useful information on both the true positive rate and the false positive rate of the test data. In contrast, this strategy is not readily applicable to MCSpu, because the false positive rate in the training data is not available due to the lack of true labels for unlabeled instances, hence, no ROC curve is obtained for the training data. Nonetheless, the threshold for MCSpu may be set based on the estimated true positive rate of the training data using the MCSpu distribution for the labeled true positives (Fig. S8).

In practice, the proposed method is particularly useful when only a small amount of labeled data is available, for instance, in sparse networks or frontier regions. By correcting a pre-trained DL model with only a limited number of labeled samples, one can build an effective earthquake detection model. Implementation of the proposed method requires model building only once, performed offline, which takes several minutes. The computational cost of training the second-stage classifier is relatively high (about 15 minutes for a sample size of 500), compared with the baseline methods (2–10 seconds for a sample size of 500; Tables S15–S17). Hence, although somewhat time-consuming, this does not hinder practical application. Furthermore, when additional labeled data become available, the model can potentially be updated by rebuilding it. Such additional data may be obtained from the detection results of the proposed method: Waveforms processed by the second-stage classifier can be labeled as true positives or false positives. Although this labeling requires seismological expertise, the detection results provided by the proposed method may offer valuable insights to facilitate the task.

Further, the proposed method can be readily integrated into operational seismic networks. The implementation procedure is as follows. First, the second-stage classification model is built in a station-wise manner offline. Next, both the first- and second-stage classification models are deployed in a station-wise implementation. This system can operate in near real time, with only a small computational latency of a few seconds, for example, when extracting waveforms from data streams every 0.1 s. A potential practical barrier, however, is the quality of the labeled training data.

In the present study, to evaluate the performance of the proposed method, we estimated the model parameters using balanced training data in which numbers of true positives and false positives are the same. On the other hand, it is known that imbalance training data may possibly bias parameter estimation, diminishing generalization capability of the model (Scrucca 2023). Parameter bias originated from data imbalance is related to other factors such as small sample size and class separability (Sun et al. 2009). Hence, it is not straightforward to discuss isolated effect of data imbalance for bias estimation. Nonetheless, it is useful to examine how the results will be affected by data imbalance in our simulation context. To this end, we carried out an additional study for Toy data 1, setting numbers of true and false positives to 20 and 2000 for training data, respectively, while the numbers for test data were kept to 100 for each label. We applied the proposed method to 100 samples of this setting. Confusion matrices for the concatenated results are summarized in Table S18, which suggests that AUC=0.96, Recall=0.91 and Precision=0.38 for the training data. Apparently, strong parameter bias occurred because of low performance of precision. On the other hand, for the test data, we obtained AUC=0.96, Recall=0.90 and Precision=0.98, which did not suggest strong estimation bias. Lastly, for the test data, we adjusted the proportion of the number of true and false positives to 1:100 as in the training data. As a result, we obtained Precision=0.37 (AUC and Recall remained the same from its definition), which yielded as low a score as that of the training data. This simple simulation study suggests that parameter estimation may not be largely affected by imbalance of training data, but there is need to carefully tune the threshold for MCS and MCSpu for imbalance test data. In addition, we also tested performance of the proposed method with extremely small number of instances. For toy data 1, we set the sample size n to five and ten. When $n = 5$, we obtained the mean test AUC=0.50 (std 0), which is a degenerated case with only one cluster yielded. Further, when $n = 10$, we obtained the mean test AUC=0.53 (std 0.10). These results suggest that $n \leq 10$ may be too small to apply the proposed method.

Finally, we discuss limitations of the present paper. First, this study did not consider S-wave detection. However, the proposed method is readily applicable for S-wave detection without any change of the algorithm, because S-wave is also thought to have a different amplitude pattern from noise that may include effect of precedent P-wave (e.g., Trnkoczy (2009)). Second, our second-stage classifier did not address the problem of minimizing the false negative cases. Because of the large sample space including various types of noise waveforms, it would be challenging to develop an appropriate classifier for negative cases using only a small number of waveforms. A possible approach would be to identify seismic waveforms among negative cases by cross correlation with known seismic waveform ('template matching', e.g., Kurihara & Obara (2021)). Alternatively, false negatives can be reduced by lowering the detection threshold of the first-stage classifier. However, in so doing, it is essential to evaluate the overall performance of the proposed method including both positive and negative cases. Third, since our second-stage classifier is based on amplitude patterns, it is expected that the classifier is little sensitive to frequency patterns of waveforms. Hence, it does not potentially discriminate between natural and non-natural earthquakes (e.g., anthropogenic bursts), given similar amplitudes patterns. Nonetheless, when frequency phases are aligned across all waveforms for one class, the classifier may be able to distinguish between true and false positives, even when the amplitude patterns are identical. As an extreme case, we carried out an additional simulation study using the similar setting of Toy data 1: sample size 20; both true and false positives are drawn by $\boldsymbol{\sigma} = (0.1, 1, 0.5)$. Subsequently, for false positive cases in the second segment, we set that amplitude sign alternates starting with positive. That is, for the jth datapoint in the second segment, $x_j \leftarrow |x_j|$ if j is odd, whereas $x_j \leftarrow -|x_j|$ if j is even. In this simulation, it is found that the classifier completely discriminated between true/false positives with a mean test AUC of 1. On the other hand, when the starting sign is randomly determined, the classifier failed to discriminate between true/false positives with a mean test AUC of 0.5. More details on the effect of phase patterns will be an interesting direction for future work. Fourth, we did not carry out sanity check (Suarez & Beroza 2024) for benchmark datasets using the proposed method. In the PU setting, one may build the second-stage model using dozens of true positive waveforms of good quality. Subsequently, applying the model to other positive waveforms may help in performing a sanity check of positive waveforms, which would be useful for cleaning benchmark datasets. Finally, we did not explore other possibilities for modeling a second-stage. Rather than Gaussian models, other models such as gamma distributions may capture useful features for waveform classification. Or, rather than raw waveform, spectral framework may be useful for better classification of waveforms. Also, we did not explore different values of the concentration parameter α to control the number of segments and clusters, but instead fixed it to one. Depending on data, other values may be appropriate for better characterization of waveforms.

5 DATA AND CODES

STEAD data are available from `https://github.com/smousavi05/STEAD`. Further, SCSN data is available from `https://scedc.caltech.edu/data/deeplearning.html`. MeSO-net data, including its catalog, are available from `https://www.eri.u-tokyo.ac.jp/project/iSeisBayes/`

MeSO-net_Narita-Array_2020/. The python codes to reproduce the results of this study are available at xxx upon acceptance.

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Supplementary Information for "Two-stage approach for earthquake detection using multiple clustering-based classification"

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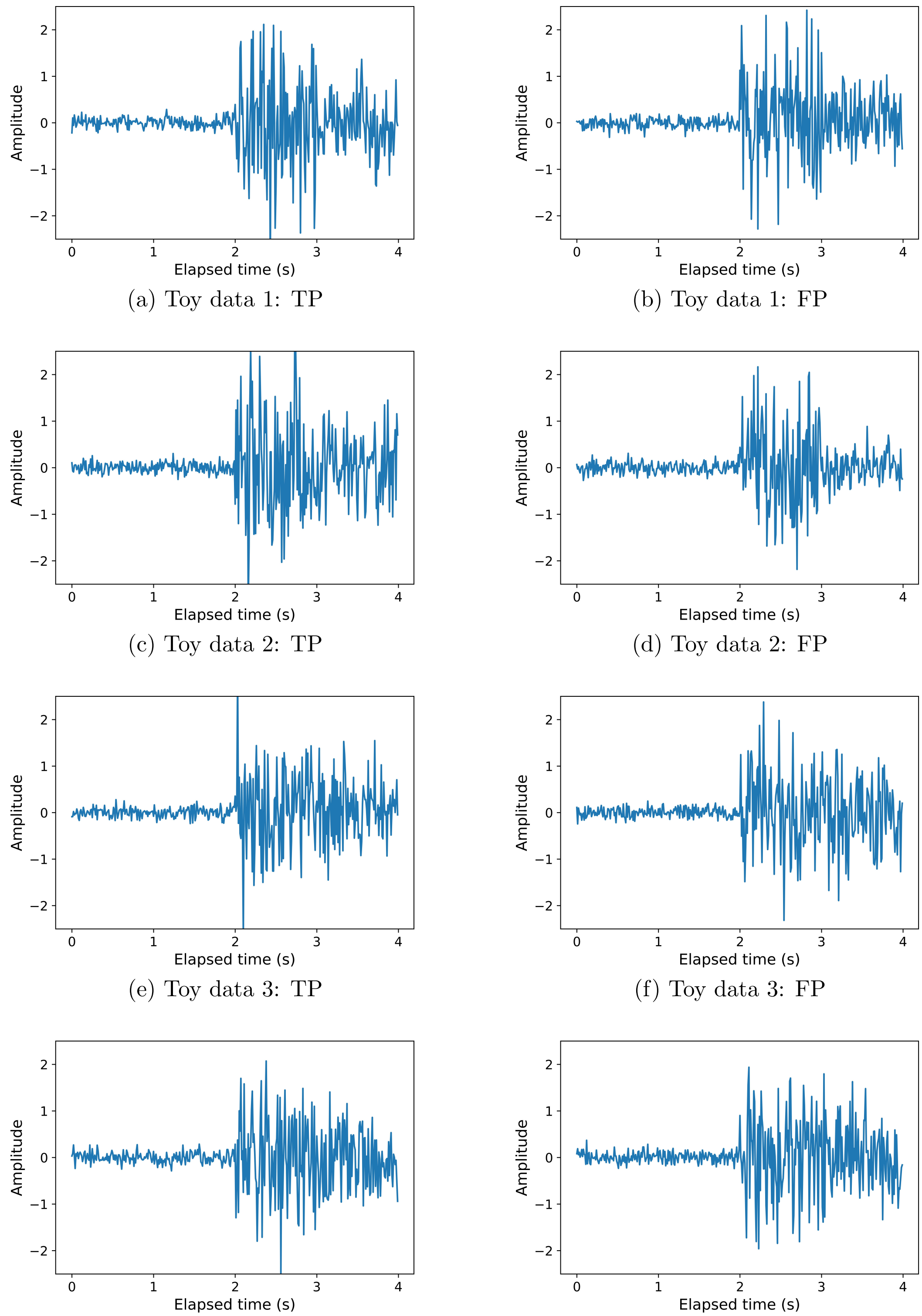


(a) Toy data 1: TP

(b) Toy data 1: FP

(c) Toy data 2: TP

(d) Toy data 2: FP

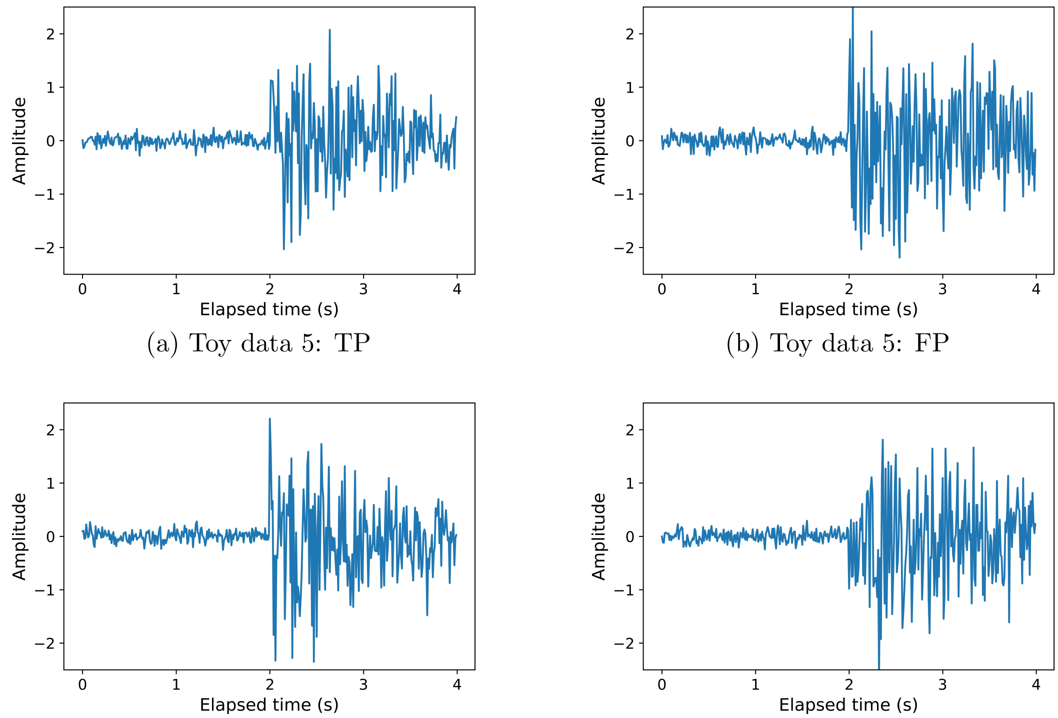
(e) Toy data 3: TP

(f) Toy data 3: FP

(g) Toy data 4: TP

(h) Toy data 4: FP

Figure S1: Waveform examples for Toy data 1-4. Panel (a): True positive (TP) of Toy data 1. Panel (b): False positive (FP) of Toy data 1. Panel (c): True positive of Toy data 2. Panel (d): False positive of Toy data 2. Panel (e): True positive of Toy data 3. Panel (f): False positive of Toy data 3. Panel (g): True positive of Toy data 4. Panel (h): False positive of Toy data 4.



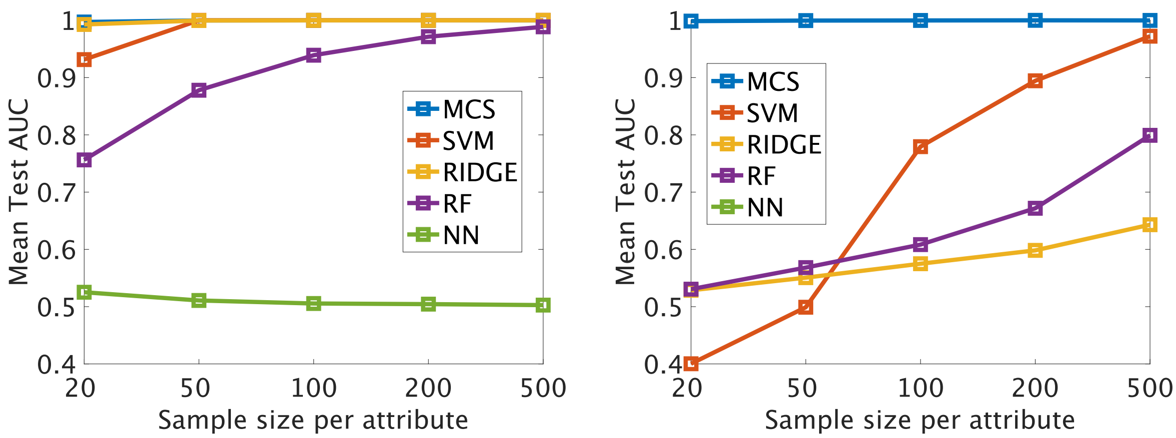
(a) Toy data 5: TP

(b) Toy data 5: FP

(c) Toy data 6: TP

(d) Toy data 6: FP

Figure S2: Waveform examples for Toy data 5-6. Panel (a): True positive (TP) of Toy data 5. Panel (b): False positive (FP) of Toy data 5. Panel (c): True positive of Toy data 6. Panel (d): False positive of Toy data 6.



(a) Toy data 5 (b) Toy data 6

Figure S3: Classification performance of true and false positives for Toy data with exponential decay of amplitude. (a) Toy data 5 (continuous, one modal). (b) Toy data 6 (continuous, two modals). For both Toy data 5 and 6, the true positive waveform exponentially decays with $\sigma = \exp\{-\lambda(t-2)\}$ in $2 \leq t \leq 4$ where $\lambda = -\log 0.4/2$. The remainder of setting was the same as Toy data 3 and 4, respectively. Note that the mean AUC for NN in (b) was below 0.4. MCS is the proposed method.

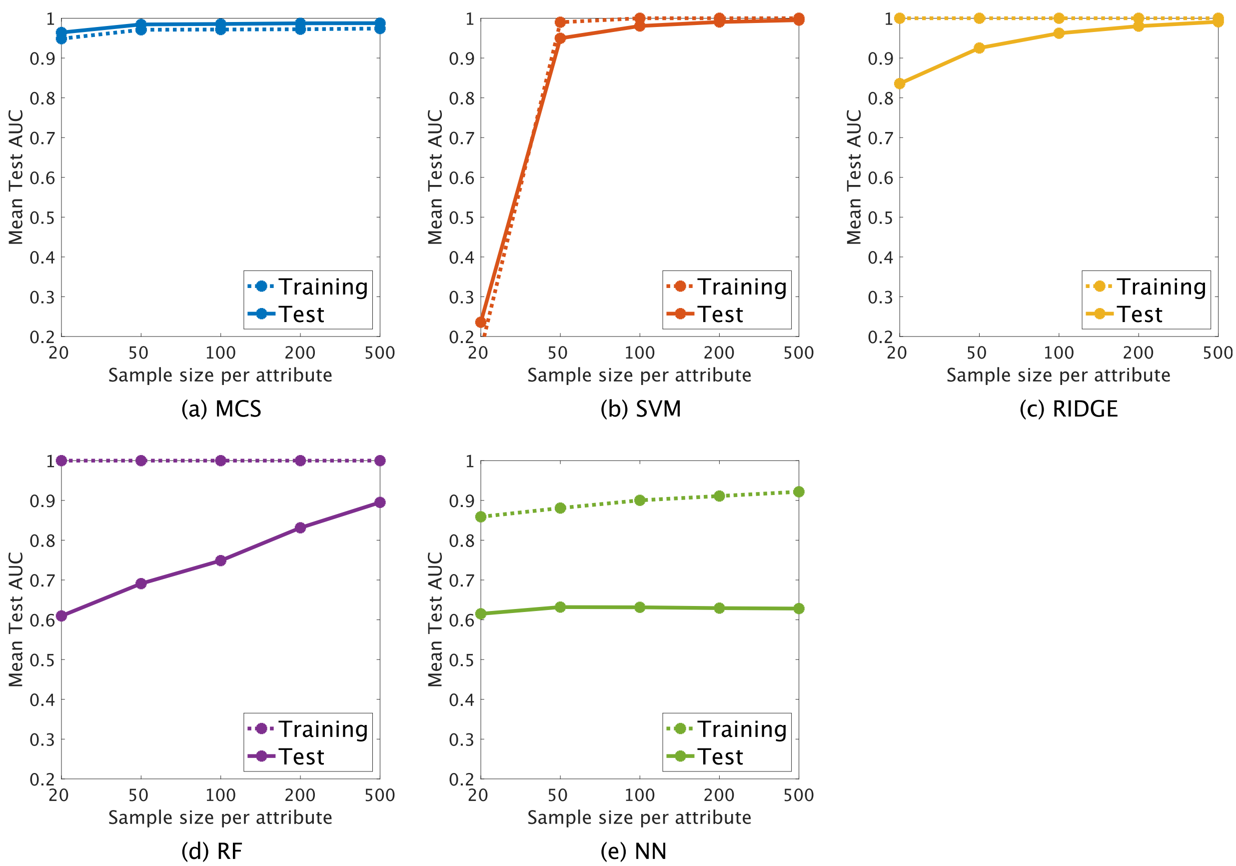
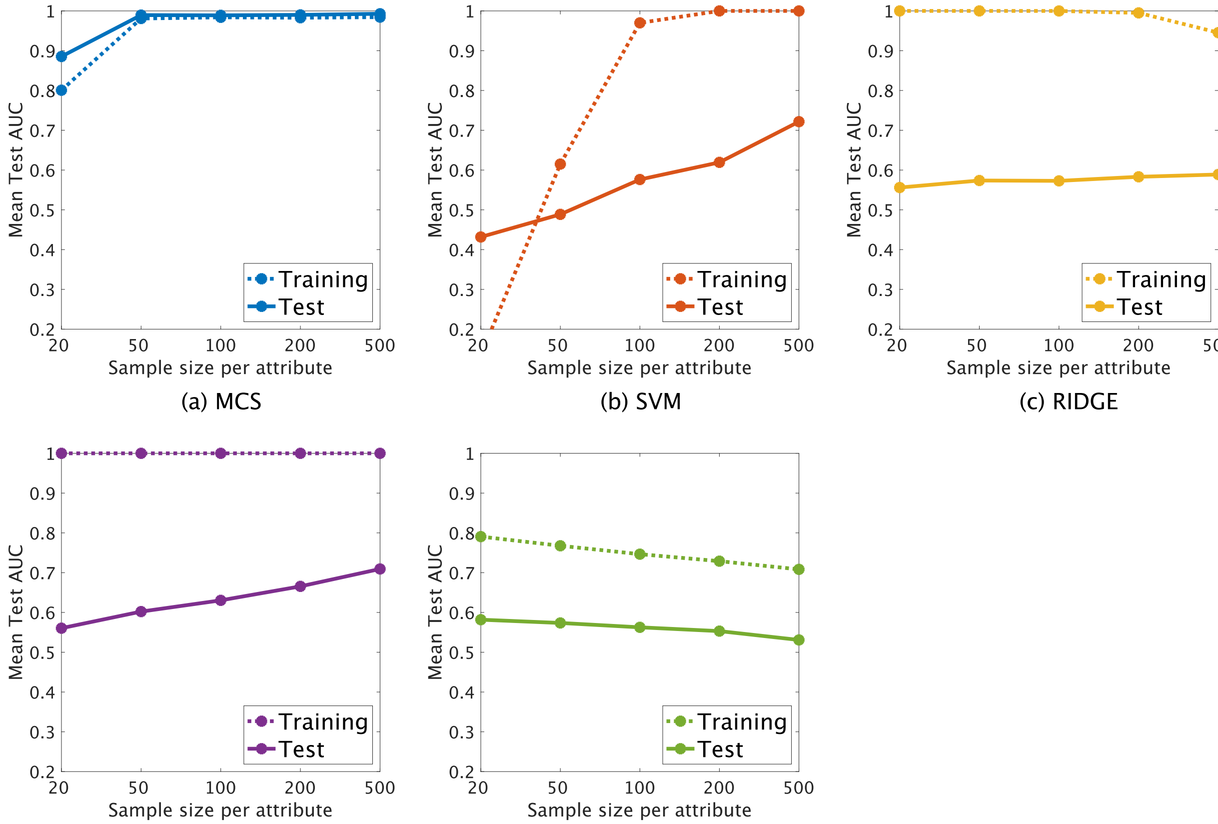
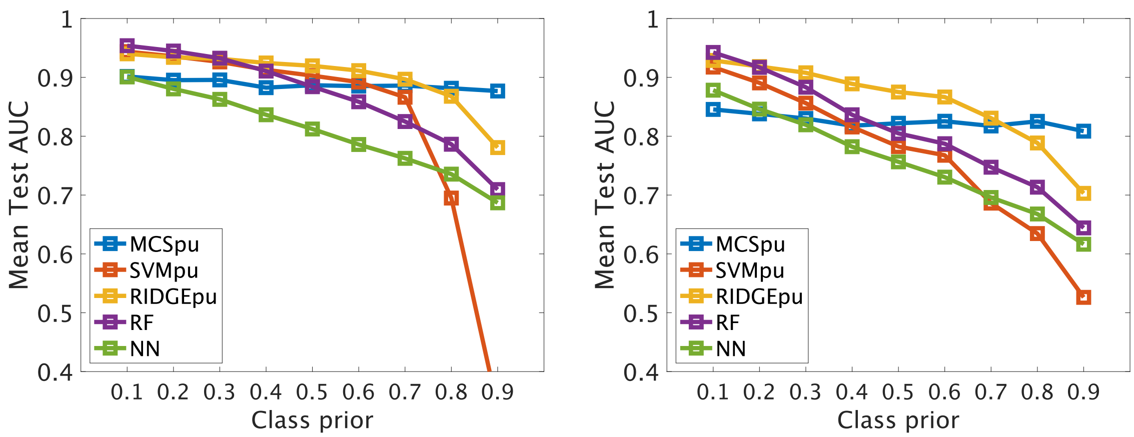


Figure S4: Classification performance of true and false positives for Toy data 1. Panels (a)-(e) for MCS, SVM, RIDGE, RF and NN, respectively. For each panel, the dotted line denotes the performance on the training data, whereas the sold line denotes the performance on the test data.



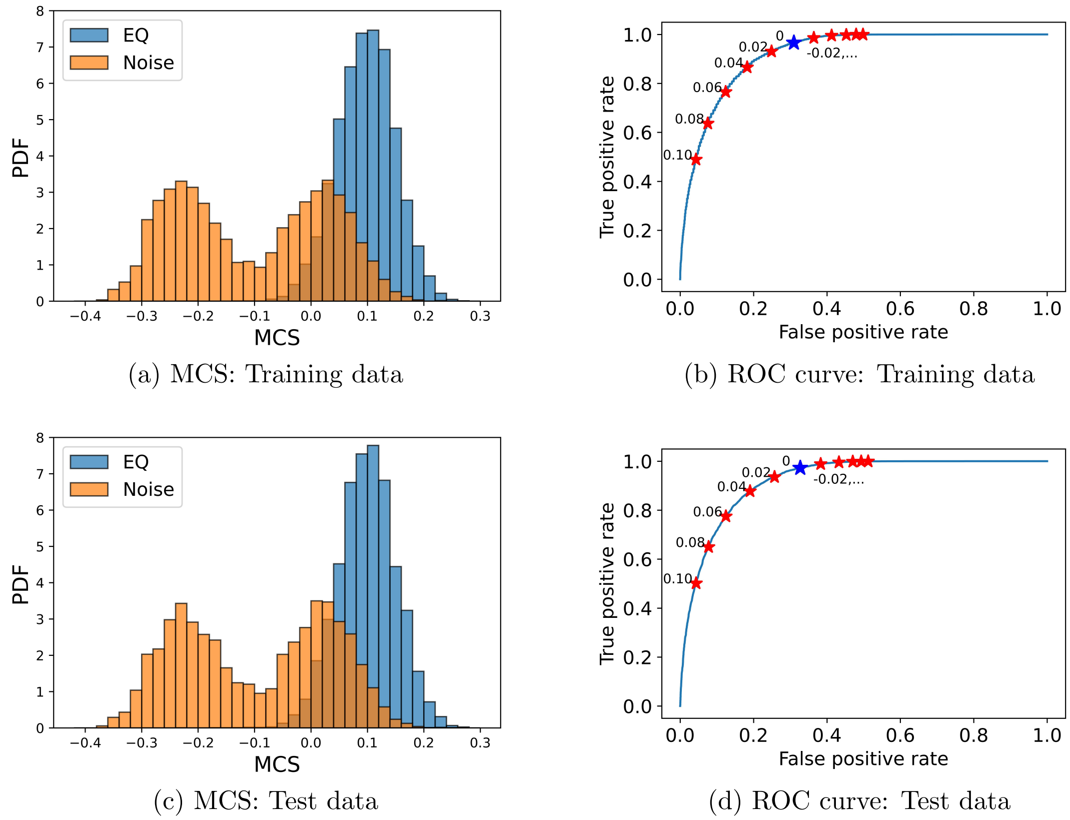
(a) MCS (b) SVM (c) RIDGE (d) RF (e) NN

Figure S5: Classification performance of true and false positives for Toy data 2. Panels (a)-(e) for MCS, SVM, RIDGE, RF and NN, respectively. For each panel, the dotted line denotes the performance on the training data, whereas the sold line denotes the performance on the test data.



(a) STEAD labeled positive (b) SCSN labeled positive

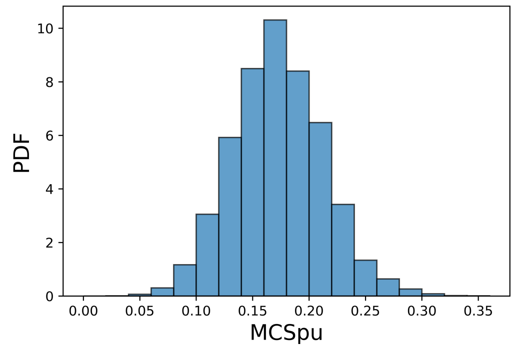
Figure S6: PU classification performance for STEAD data. (a) case (1), and (b) case (2). The proposed method was based on MCSpu in Eq.(5) of the main manuscript ('MCSpu'). SVM and RIDGE were based on Empirical Risk Minimization (denoted as 'SVMpu', 'RIDGEpu' in those panels). RF was based on the biased learning framework, whereas for NN, only labeled instances were considered as positive. Note that the results other than SVMpu and RIDGEpu are identical to those in Fig.6 of the main manuscript. For details of results, please refer to Appendix Tables S13 and S14.



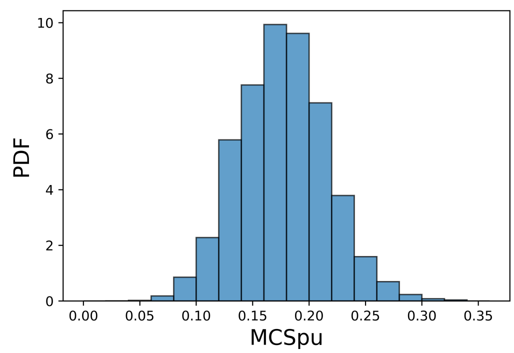
(a) MCS: Training data (b) ROC curve: Training data

(c) MCS: Test data (d) ROC curve: Test data

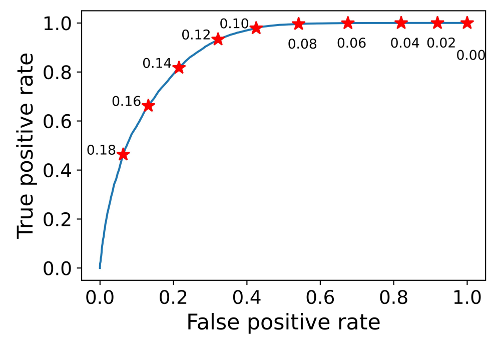
Figure S7: Distribution of MCS and ROC curve for STEAD data. Panels (a) and (b) are for the training data, whereas Panels (c) and (d) for the test data of STEAD data analysis in Section 3.2 of the main manuscript. We concatenated all datasets generated for Fig. 5a. An asterisk in Panels (b) and (d) denotes a pair of true and false positive rates for a given threshold of MCS shown in text. The asterisk in blue is for the default setting of threshold 0, whereas that in red for the other settings. It is observed that the distribution of MCS and ROC curve are similar between the training and test data.



(a) MCSpu: Trarining



(b) MCSpu: Test data



(c) ROC curve: Test data

Figure S8: Distribution of MCSpu and ROC curve for STEAD data. Panel (a): Distribution of MCSpu for labeled true positives of the training data. Panels (b) and (c): Distribution of MCSpu for true positives and ROC curve for the test data. Both training and test data are from STEAD data for the PU classification in Section 3.3 of the main manuscript. We concatenated all datasets for class prior 0.5. An asterisk in Panel (c) denotes a pair of true and false positive rates for a given threshold of MCSpu shown in text. Since the class prior is not known in advance, no ROC curve is obtained for the training data. It is observed that the distribution of MCSpu is similar between the training and test data.

Table S1: Toy data 1 classification performance. For each method, the first row denotes mean of AUC over 100 samples, whereas the second row standard deviation of AUC.

Sample size per attribute	20	50	100	200	500
MCS mean	0.96	0.98	0.99	0.99	0.99
std	(0.08)	(0.01)	(0.01)	(0.01)	(0.01)
SVM mean	0.24	0.95	0.98	0.99	1.00
std	(0.26)	(0.09)	(0.01)	(0.01)	(0.00)
RIDGE mean	0.84	0.93	0.96	0.98	0.99
std	(0.03)	(0.02)	(0.01)	(0.01)	(0.00)
RF mean	0.61	0.69	0.75	0.83	0.90
std	(0.05)	(0.04)	(0.03)	(0.03)	(0.02)
NN mean	0.62	0.63	0.63	0.63	0.63
std	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)

Table S2: Toy data 2 classification performance. For each method, the first row denotes mean of AUC over 100 samples, whereas the second row standard deviation of AUC.

Sample size per attribute	20	50	100	200	500
MCS mean	0.89	0.99	0.99	0.99	0.99
std	(0.12)	(0.01)	(0.01)	(0.01)	(0.01)
SVM mean	0.43	0.49	0.58	0.62	0.72
std	(0.07)	(0.10)	(0.06)	(0.07)	(0.05)
RIDGE mean	0.56	0.57	0.57	0.58	0.59
std	(0.05)	(0.04)	(0.04)	(0.05)	(0.04)
RF mean	0.56	0.60	0.63	0.67	0.71
std	(0.05)	(0.04)	(0.05)	(0.04)	(0.04)
NN mean	0.58	0.57	0.56	0.55	0.53
std	(0.04)	(0.04)	(0.04)	(0.05)	(0.04)

Table S3: Toy data 3 classification performance. For each method, the first row denotes mean of AUC over 100 samples, whereas the second row standard deviation of AUC.

Sample size per attribute	20	50	100	200	500
MCS mean	0.91	0.99	0.99	0.99	0.99
std	(0.14)	(0.01)	(0.01)	(0.01)	(0.00)
SVM mean	0.36	0.99	1.00	1.00	1.00
std	(0.39)	(0.01)	(0.00)	(0.00)	(0.00)
RIDGE mean	0.89	0.96	0.98	0.99	1.00
std	(0.03)	(0.01)	(0.01)	(0.00)	(0.00)
RF mean	0.63	0.73	0.79	0.85	0.90
std	(0.05)	(0.04)	(0.04)	(0.03)	(0.02)
NN mean	0.57	0.54	0.53	0.52	0.51
std	(0.03)	(0.02)	(0.02)	(0.01)	(0.01)

Table S4: Toy data 4 classification performance. For each method, the first row denotes mean of AUC over 100 samples, whereas the second row standard deviation of AUC.

Sample size per attribute	20	50	100	200	500
MCS mean	0.81	0.99	1.00	1.00	1.00
std	(0.10)	(0.01)	(0.01)	(0.01)	(0.01)
SVM mean	0.45	0.51	0.57	0.65	0.84
std	(0.08)	(0.09)	(0.06)	(0.07)	(0.04)
RIDGE mean	0.56	0.57	0.57	0.58	0.59
std	(0.06)	(0.04)	(0.04)	(0.05)	(0.04)
RF mean	0.57	0.60	0.63	0.67	0.72
std	(0.05)	(0.05)	(0.04)	(0.04)	(0.04)
NN mean	0.38	0.39	0.40	0.41	0.41
std	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)

Table S5: STEAD classification performance for $w = 0$. For each method, the first row denotes mean of AUC over 100 samples, whereas the second row standard deviation of AUC.

Sample size per attribute	20	50	100	200	500
MCS mean	0.89	0.92	0.93	0.93	0.94
std	(0.04)	(0.02)	(0.02)	(0.02)	(0.02)
SVM mean	0.94	0.95	0.96	0.96	0.97
std	(0.02)	(0.02)	(0.01)	(0.01)	(0.01)
RIDGE mean	0.91	0.93	0.94	0.94	0.95
std	(0.03)	(0.02)	(0.02)	(0.01)	(0.02)
RF mean	0.93	0.94	0.95	0.96	0.96
std	(0.03)	(0.02)	(0.02)	(0.01)	(0.01)
NN mean	0.88	0.91	0.92	0.93	0.94
std	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)

Table S6: STEAD classification performance for $w = 0.1$. For each method, the first row denotes mean of AUC over 100 samples, whereas the second row standard deviation of AUC.

Sample size per attribute	20	50	100	200	500
MCS mean	0.91	0.93	0.94	0.94	0.95
std	(0.03)	(0.02)	(0.02)	(0.02)	(0.01)
SVM mean	0.95	0.96	0.96	0.97	0.97
std	(0.02)	(0.01)	(0.01)	(0.01)	(0.01)
RIDGE mean	0.90	0.92	0.92	0.93	0.94
std	(0.04)	(0.02)	(0.02)	(0.02)	(0.02)
RF mean	0.93	0.95	0.95	0.96	0.97
std	(0.03)	(0.01)	(0.02)	(0.01)	(0.01)
NN mean	0.88	0.90	0.91	0.93	0.94
std	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)

Table S7: STEAD classification performance for $w = 0.2$. For each method, the first row denotes mean of AUC over 100 samples, whereas the second row standard deviation of AUC.

Sample size per attribute	20	50	100	200	500
MCS mean	0.91	0.94	0.95	0.95	0.96
std	(0.03)	(0.02)	(0.01)	(0.01)	(0.01)
SVM mean	0.95	0.97	0.97	0.97	0.98
std	(0.02)	(0.01)	(0.01)	(0.01)	(0.01)
RIDGE mean	0.88	0.91	0.91	0.92	0.93
std	(0.04)	(0.03)	(0.03)	(0.02)	(0.02)
RF mean	0.93	0.95	0.96	0.96	0.97
std	(0.03)	(0.02)	(0.01)	(0.01)	(0.01)
NN mean	0.87	0.90	0.91	0.92	0.93
std	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)

Table S8: STEAD classification performance for $w = 0.3$. For each method, the first row denotes mean of AUC over 100 samples, whereas the second row standard deviation of AUC.

Sample size per attribute	20	50	100	200	500
MCS mean	0.92	0.95	0.96	0.96	0.97
std	(0.03)	(0.02)	(0.01)	(0.01)	(0.01)
SVM mean	0.96	0.97	0.98	0.98	0.99
std	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
RIDGE mean	0.86	0.89	0.89	0.91	0.91
std	(0.05)	(0.03)	(0.03)	(0.02)	(0.02)
RF mean	0.93	0.95	0.96	0.96	0.97
std	(0.03)	(0.02)	(0.01)	(0.01)	(0.01)
NN mean	0.87	0.89	0.90	0.91	0.92
std	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)

Table S9: STEAD classification performance for $w = 0.7$. For each method, the first row denotes mean of AUC over 100 samples, whereas the second row standard deviation of AUC.

Sample size per attribute	20	50	100	200	500
MCS mean	0.86	0.91	0.93	0.93	0.94
std	(0.05)	(0.03)	(0.02)	(0.02)	(0.02)
SVM mean	0.86	0.92	0.93	0.95	0.96
std	(0.13)	(0.02)	(0.02)	(0.01)	(0.01)
RIDGE mean	0.69	0.73	0.75	0.77	0.78
std	(0.06)	(0.05)	(0.04)	(0.03)	(0.04)
RF mean	0.89	0.92	0.93	0.95	0.96
std	(0.04)	(0.02)	(0.02)	(0.02)	(0.01)
NN mean	0.69	0.75	0.77	0.80	0.83
std	(0.05)	(0.04)	(0.04)	(0.03)	(0.03)

Table S10: STEAD classification performance for $w = 0.8$. For each method, the first row denotes mean of AUC over 100 samples, whereas the second row standard deviation of AUC.

Sample size per attribute	20	50	100	200	500
MCS mean	0.85	0.89	0.91	0.92	0.92
std	(0.05)	(0.03)	(0.02)	(0.02)	(0.02)
SVM mean	0.67	0.84	0.87	0.90	0.92
std	(0.21)	(0.04)	(0.03)	(0.02)	(0.02)
RIDGE mean	0.61	0.64	0.66	0.68	0.69
std	(0.06)	(0.06)	(0.05)	(0.04)	(0.05)
RF mean	0.88	0.90	0.92	0.93	0.94
std	(0.04)	(0.02)	(0.02)	(0.02)	(0.02)
NN mean	0.60	0.66	0.68	0.72	0.76
std	(0.05)	(0.04)	(0.04)	(0.04)	(0.03)

Table S11: STEAD classification performance for $w = 0.9$. For each method, the first row denotes mean of AUC over 100 samples, whereas the second row standard deviation of AUC.

Sample size per attribute	20	50	100	200	500
MCS mean	0.82	0.86	0.88	0.89	0.89
std	(0.06)	(0.03)	(0.03)	(0.02)	(0.02)
SVM mean	0.50	0.57	0.68	0.75	0.82
std	(0.10)	(0.12)	(0.06)	(0.04)	(0.03)
RIDGE mean	0.53	0.54	0.55	0.57	0.57
std	(0.05)	(0.05)	(0.04)	(0.04)	(0.05)
RF mean	0.86	0.89	0.91	0.91	0.92
std	(0.04)	(0.02)	(0.02)	(0.02)	(0.02)
NN mean	0.53	0.55	0.56	0.59	0.64
std	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)

Table S12: STEAD classification performance for $w = 1$. For each method, the first row denotes mean of AUC over 100 samples, whereas the second row standard deviation of AUC.

Sample size per attribute	20	50	100	200	500
MCS mean	0.50	0.50	0.51	0.50	0.50
std	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
SVM mean	0.50	0.50	0.51	0.51	0.50
std	(0.03)	(0.04)	(0.04)	(0.04)	(0.04)
RIDGE mean	0.49	0.50	0.50	0.51	0.49
std	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
RF mean	0.50	0.50	0.50	0.51	0.50
std	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
NN mean	0.50	0.50	0.50	0.50	0.50
std	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)

Table S13: PU classification performance for STEAD labeled positive. For each method, the first row denotes mean of AUC over 100 samples, whereas the second row standard deviation of AUC.

Class prior	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
MCSpu mean	0.90	0.90	0.90	0.88	0.89	0.89	0.89	0.88	0.88
std	(0.03)	(0.03)	(0.03)	(0.03)	(0.04)	(0.03)	(0.03)	(0.04)	(0.04)
SVM mean	0.96	0.95	0.92	0.88	0.85	0.83	0.79	0.71	0.58
std	(0.01)	(0.02)	(0.02)	(0.03)	(0.04)	(0.04)	(0.05)	(0.16)	(0.19)
SVMpu mean	0.94	0.94	0.93	0.91	0.90	0.89	0.87	0.69	0.34
std	(0.02)	(0.02)	(0.02)	(0.03)	(0.04)	(0.04)	(0.11)	(0.30)	(0.26)
RIDGE mean	0.94	0.94	0.93	0.92	0.92	0.91	0.89	0.87	0.79
std	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)	(0.04)	(0.08)
RIDGEpu mean	0.94	0.93	0.93	0.92	0.92	0.91	0.90	0.87	0.78
std	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.02)	(0.03)	(0.04)	(0.09)
RF mean	0.95	0.94	0.93	0.91	0.88	0.86	0.82	0.79	0.71
std	(0.01)	(0.02)	(0.02)	(0.02)	(0.04)	(0.04)	(0.04)	(0.04)	(0.05)
NN mean	0.90	0.88	0.86	0.84	0.81	0.79	0.76	0.74	0.69
std	(0.02)	(0.03)	(0.03)	(0.03)	(0.04)	(0.04)	(0.04)	(0.04)	(0.05)

Table S14: PU classification performance for SCSN labeled positive. For each method, the first row denotes mean of AUC over 100 samples, whereas the second row standard deviation of AUC.

Class prior	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
MCSpu mean	0.85	0.84	0.83	0.82	0.82	0.83	0.82	0.83	0.81
std	(0.04)	(0.05)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
SVM mean	0.95	0.90	0.85	0.79	0.75	0.72	0.68	0.64	0.58
std	(0.02)	(0.03)	(0.04)	(0.04)	(0.05)	(0.05)	(0.05)	(0.05)	(0.06)
SVMpu mean	0.92	0.89	0.86	0.82	0.78	0.77	0.69	0.63	0.53
std	(0.03)	(0.04)	(0.04)	(0.05)	(0.05)	(0.06)	(0.08)	(0.08)	(0.08)
RIDGE mean	0.93	0.91	0.90	0.88	0.86	0.85	0.81	0.78	0.70
std	(0.02)	(0.03)	(0.03)	(0.03)	(0.04)	(0.04)	(0.04)	(0.06)	(0.08)
RIDGEpu mean	0.93	0.92	0.91	0.89	0.88	0.87	0.83	0.79	0.70
std	(0.02)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.04)	(0.06)	(0.08)
RF mean	0.94	0.92	0.88	0.84	0.80	0.79	0.75	0.71	0.64
std	(0.02)	(0.03)	(0.03)	(0.04)	(0.05)	(0.04)	(0.04)	(0.05)	(0.05)
NN mean	0.88	0.85	0.82	0.78	0.76	0.73	0.70	0.67	0.62
std	(0.03)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.05)	(0.05)	(0.06)

Table S15: Training time for Toy data (1–4). The training time (second) for building a model is summarized as a function of sample size per attribute. Those are averaged over four types of Toy data and 100 samples. The total computation times (second) over four data types, five sample sizes per attribute, and 100 samples are shown in the column 'Total'.

Sample size per attribute	20	50	100	200	500	Total
MCS	84.66	144.15	240.13	473.74	1017.09	783910
SVM	0.01	0.04	0.14	0.51	4.21	1961
RIDGE	0.18	0.24	0.32	0.48	1.18	964
RF	0.31	0.69	1.35	2.72	7.17	4895
NN	0.00	0.02	0.06	0.24	1.64	785

Table S16: Training time for STEAD data. The training time (second) for building a model is summarized as a function of sample size per attribute. Those are averaged over weight w (0 to 1 with the increment 0.1) and 100 samples. The total computation times (second) over 11 weights, five sample sizes per attribute and 100 samples are shown in the column 'Total'.

Sample size per attribute	20	50	100	200	500	Total
MCS	68.66	135.28	259.43	455.48	1067.49	2184980
SVM	0.01	0.04	0.11	0.35	2.58	3392
RIDGE	0.33	0.61	1.15	2.13	9.80	15433
RF	0.31	0.67	1.31	2.65	7.03	13172
NN	0.00	0.02	0.06	0.26	1.82	2378

Table S17: Training time for PU classification. The training time (second) for building a model is summarized as a function of class prior. Those are averaged over two settings of data (STEAD labeled positive and SCSN labeled positive) and 100 samples. The total computation times (second) over nine class priors, two cases of data and 100 samples are shown in the column 'Total'.

Class prior	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	Total
MCSpu	562.86	564.12	563.54	563.51	565.35	583.25	594.38	590.10	604.74	519180
SVM	0.52	0.56	0.61	0.69	0.78	0.81	0.84	0.88	0.89	658
RIDGE	2.29	2.32	2.34	2.30	2.28	2.27	2.23	2.23	2.22	2047
RF	3.17	3.20	3.24	3.29	3.34	3.41	3.47	3.54	3.62	3028
NN	0.43	0.43	0.43	0.43	0.42	0.42	0.42	0.42	0.42	384

Table S18: Performance results by the proposed method for imbalance instances of Toy data 1. The confusion matrix (a) summarizes results of 100 samples for the training data, setting the numbers of true and false positives to 20 and 2000, respectively. Here, TP and FP denote true and false positives, respectively. On the other hand, the confusion matrix (b) summarizes results of 100 samples for the test data, setting the numbers of true and false positives to 100 and 100, respectively. For both cases, we set the classification threshold of MCS to zero. From the confusion matrix (a), we obtained AUC = 0.96, Recall=0.91 and Precision = 0.38. On the other hand, from the confusion matrix (b), we obtained AUC = 0.96, Recall=0.90 and Precision = 0.98. Lastly, for the confusion matrix (b), we adjusted the proportion of the number of true and false positives to 1:100 as in the training data, by multiplying the second raw (i.e., the row of FP) of the confusion matrix (b) by 100. As a result of this manipulation, Precision became 0.37, whereas Recall and AUC remain the same (from its definition).

(a) Training data

		Predicted label		
		TP	FP	Total
True label	TP	1817	183	2000
	FP	2963	197037	200000
	Total	4780	197220	202000

(b) Test data

		Predicted label		
		TP	FP	Total
True label	TP	9000	1000	10000
	FP	143	9857	10000
	Total	9143	10857	20000